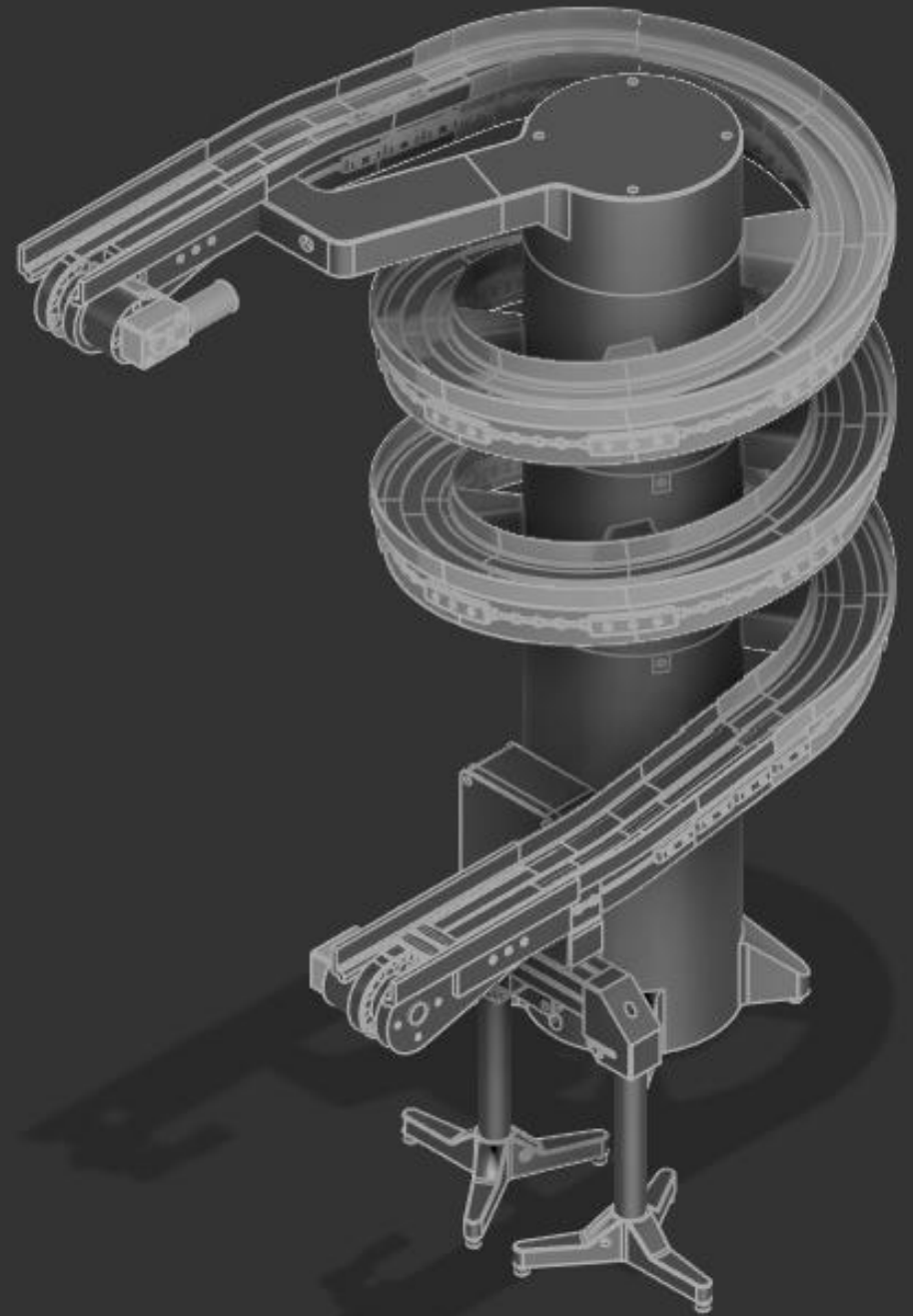


Helix Conveyor KE 50

Assembly manual



Content

01

Description of the device

02

Components for building
flexible, modular conveyors

03

Building descriptions

04

Drawings

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Links to purchasing external
parts

01-Description of the device

This is an instruction of how to 3d-print and build the components needed to build a Helix conveyor in the KE 50 flexible chain conveyor family.

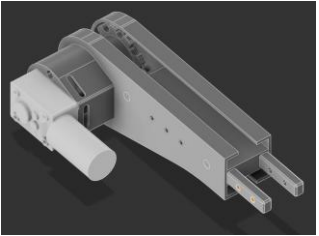
This type of conveyor is common in the manufacturing industry. Mostly, a helix is used as an elevator (up or down does not matter) but with suitable control system it can also be used as an accumulator. The small footprint and the fact that this is a continuous flow elevator, makes it very popular.

The commercial versions of helix conveyors have a ball bearing in each conveyor link to reduce friction, but in this case there is no ball bearing. This limits the number of rotations to 2.5 before the friction becomes too high (for these motors).

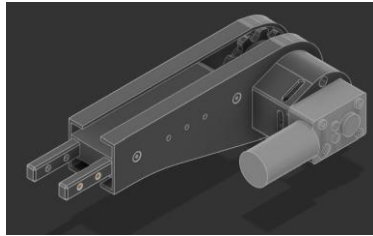
The stepmodel of the entire assembly, KE-0400.stp, is submitted among the other files. Easy way to see how everything is assembled for those that have a CAD system.

This particular conveyor is suitable for school projects or prototypes for automation ideas. Also suitable for home projects of various kinds.

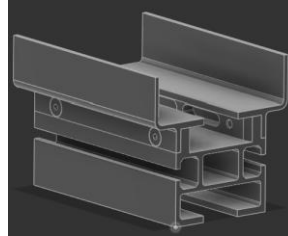
02-Components for building the Helix



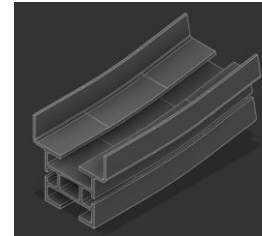
KE-0002
Drive Unit, Left



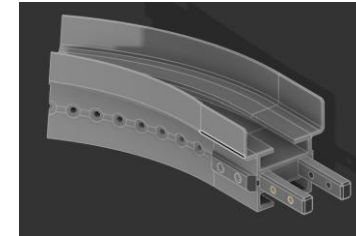
KE-0003
Drive Unit, Right



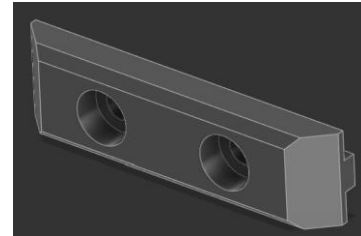
KE-0413
Chain Insertion
Section



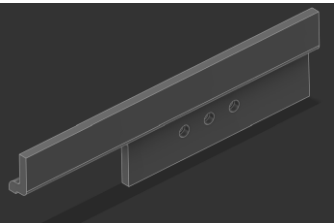
KE-0408
Vertical Bend 9
degrees, up



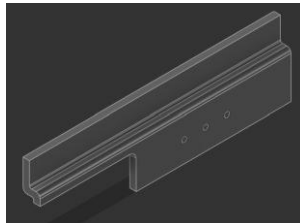
KE-0210
Helix lower end
section



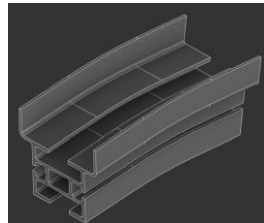
KE-0051
Clamp, Short



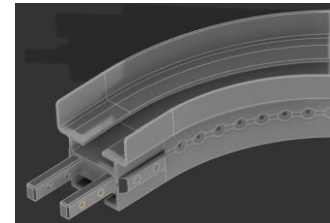
KE-0420
Guide rail for drive
unit, Left



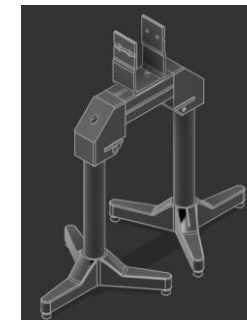
KE-0421
Guide rail for drive
unit, Left



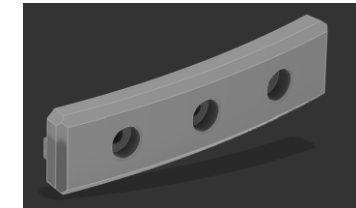
KE-0409
Vertical Bend, 9
degrees, down



KE-0212
Helix upper end
section

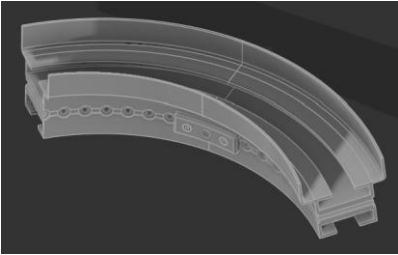


KE-0417
Leg support

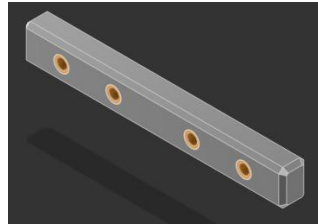


KE-0213
Connection Strip

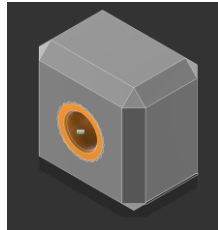
...components for building modular conveyors



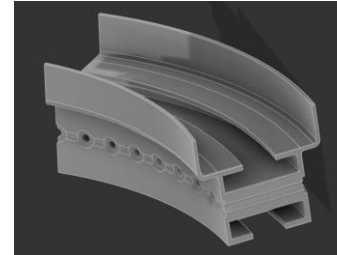
KE-0419
Helix 90 degree section



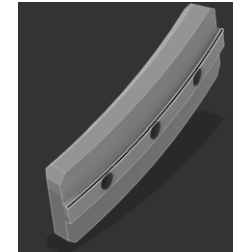
KE-0050
Connecting Rod



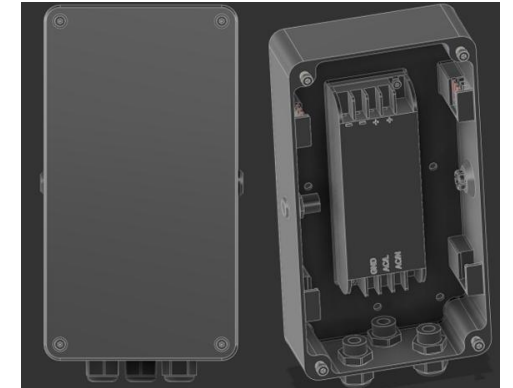
KE-0053
Slot Nut



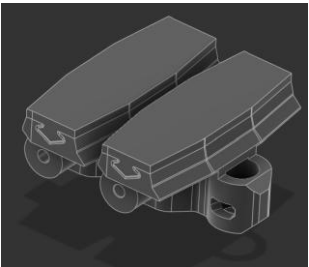
KE-0211
Helix 45 degree section



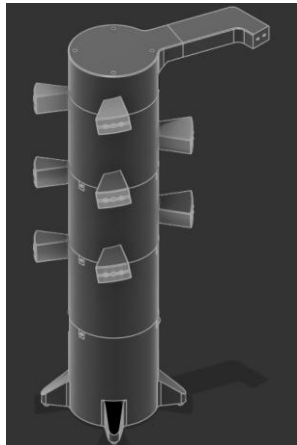
KE-0215
Connecting strip, inner



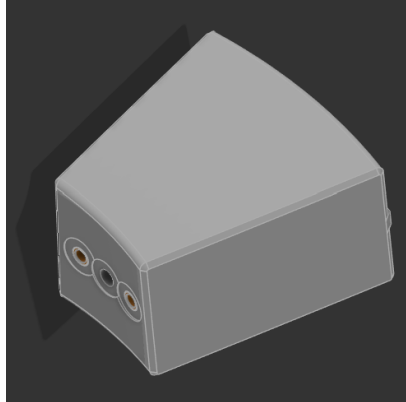
KE-0175
Electrical Cabinet. See separate
print profile and assembly
manual here:
[https://makerworld.com/en/mod
els/2052857-electrical-cabinet-
for-110-220v-ac-to-12v-dc-ke-
50#profileId-2215802](https://makerworld.com/en/models/2052857-electrical-cabinet-for-110-220v-ac-to-12v-dc-ke-50#profileId-2215802)



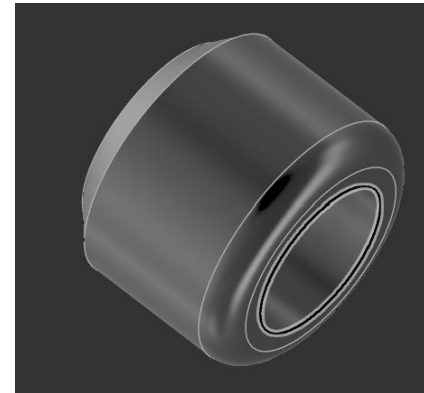
KE-0149
Friction Top Chain



KE-0401
Frame for helix



KE-0406
Conveyor bracket

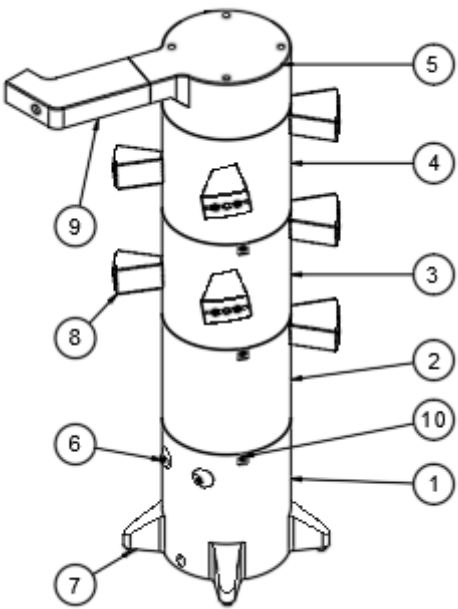


KE-0026
TPU Foot

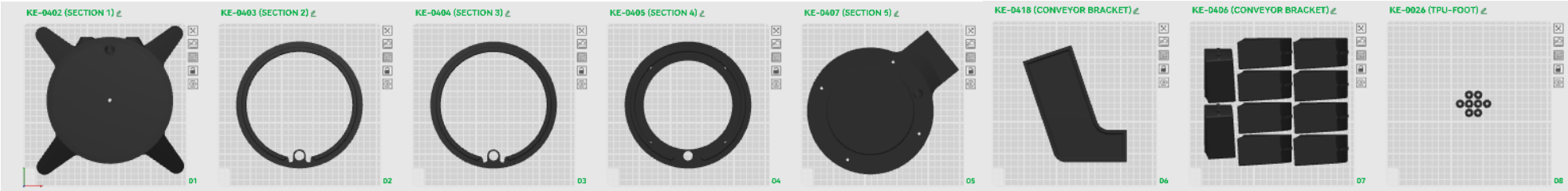
03-Assembly of the Helix frame KE-0401

Begin with the frame. When the frame is ready, it ´s easier to fit the helix conveyor.
The frame has to be built from the bottom and up.
Begin with section 1, then 2,3,4 and 5.

Follow the step by step description as follows. The material needed is shown in the BOM to the right.
Print the parts on plate 1,2,3,4,5,6,7, & 8 of the print profile KE-400 (Helix conveyor).3mf.
Everything is printed I PETG except the feet that are made of TPU.

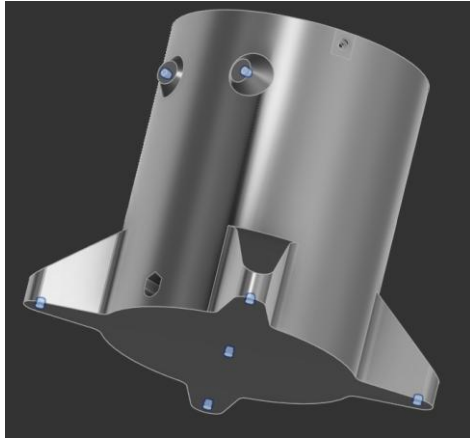


11	2	ISO 4782 - M4 x 40 Steel 4.8 Plain	Hexagon Socket Head Cap Screw ISO 4782 - M4 x 40
10	40	ISO 4782 - M4 x 12 Steel 4.8 Plain	Hexagon Socket Head Cap Screw ISO 4782 - M4 x 12
9	1	KE-0418	CONVEYOR BRACKET, TOP, Frame for helix
8	10	KE-0408	CONVEYOR BRACKET
7	5	KE-0028	TPU FOOT
6	24	AB018	Threaded insert M4
5	1	KE-0407	SECTION 5, Frame for helix
4	1	KE-0405	SECTION 4, Frame for helix
3	1	KE-0404	SECTION 3, Frame for helix
2	1	KE-0403	SECTION 2, Frame for helix
1	1	KE-0402	SECTION 1, Frame for helix
Item	Qty	Part Number	Description

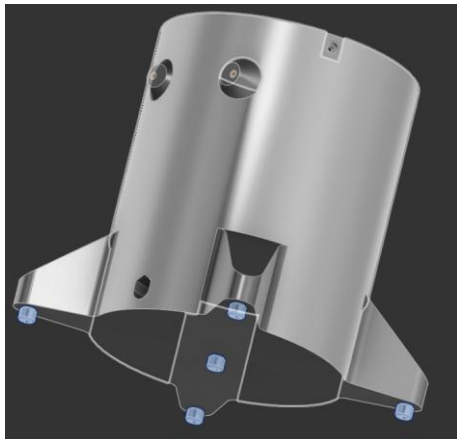


03-Assembly of the Helix frame KE-0401

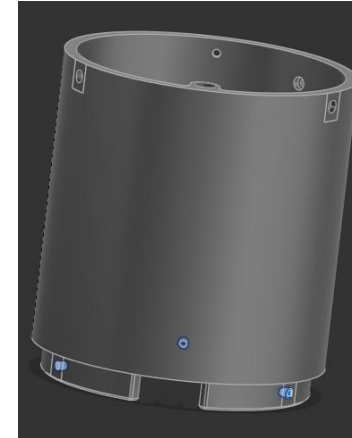
1. Put the threaded M4 inserts into Section 1, KE-0402 (7x).



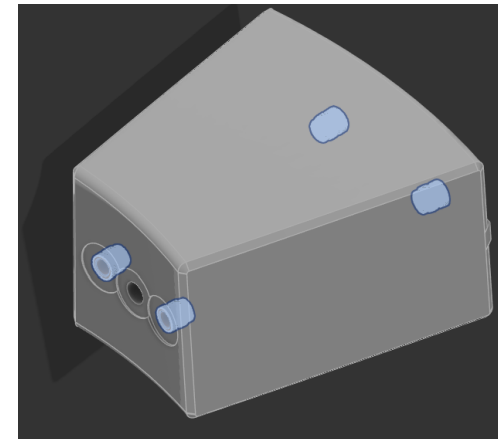
2. Bolt the TPU feet, KE-0026, using MC6S 4x12 (5x).



3. Put the M4 inserts into KE-0403 (3x).

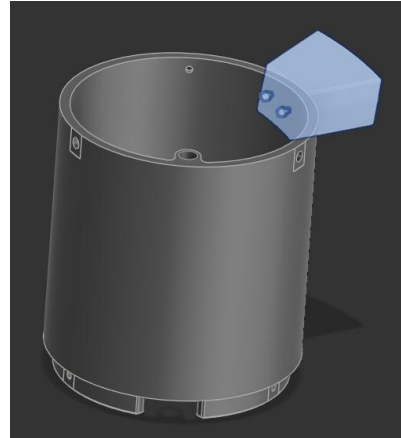


4. Put the M4 threaded inserts in KE-0406 (4x).



03-Assembly of the Helix frame KE-0401

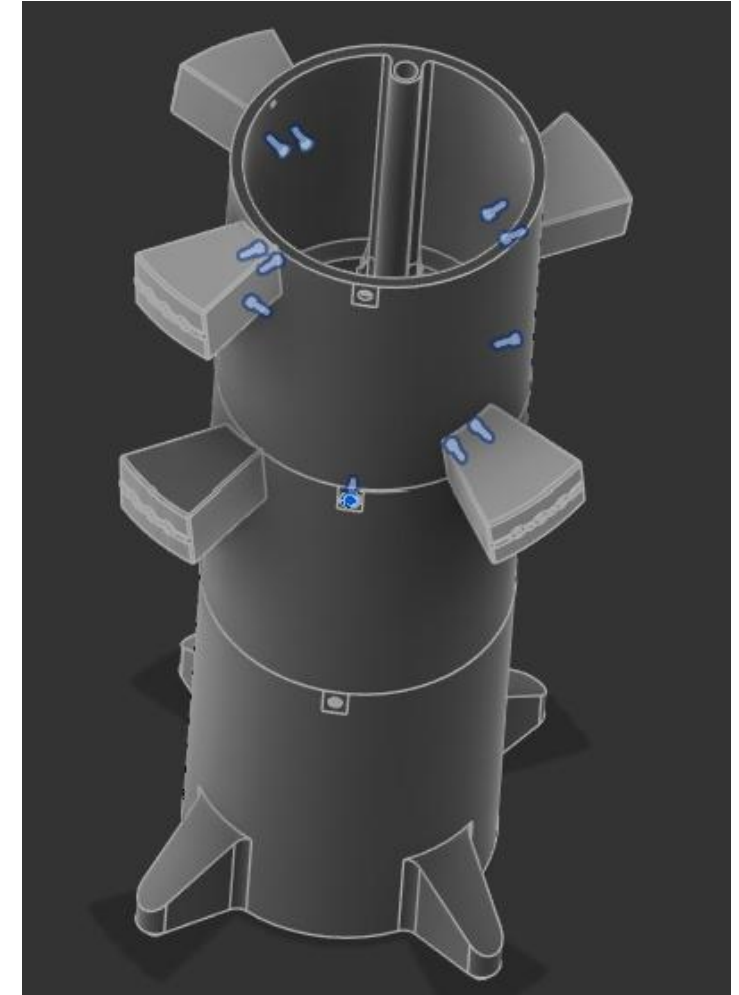
5. Bolt KE-0406 to KE-0403 from the inside using MC6S 4x12 (2x).



6. Bolt KE-0403 on to KE-0402 using MC6S 4x12 (3x). Make sure to align the cable duct (the vertical, cylindrical pipe) throughout the sections.

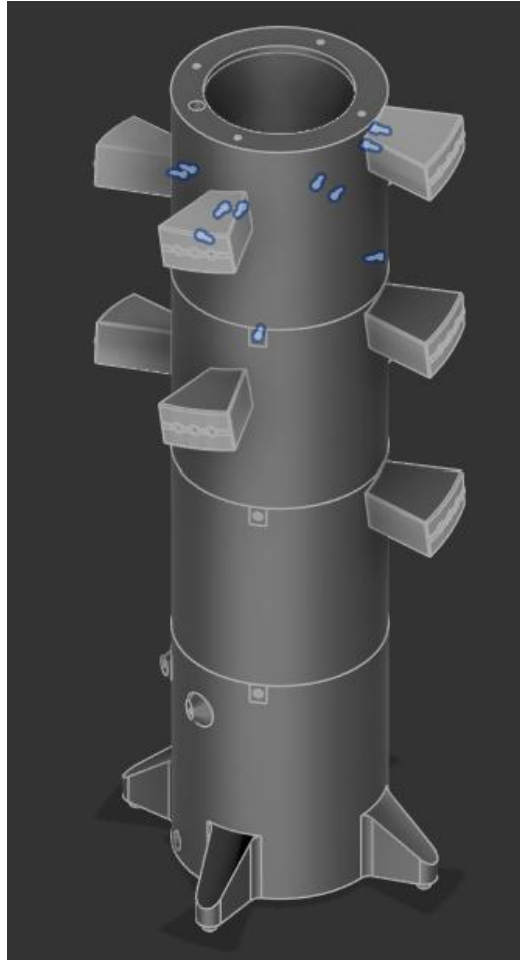


7. Put the threaded inserts in KE-404 (3x in the flange that goes into KE-403). Bolt KE-0404 on KE-0403 using MC6S (3x). Bolt KE-0406 (4x) on KE-0404 from the inside of the "tube" using MC6S 4x12 (8x). Align the cable duct with the previous sections.



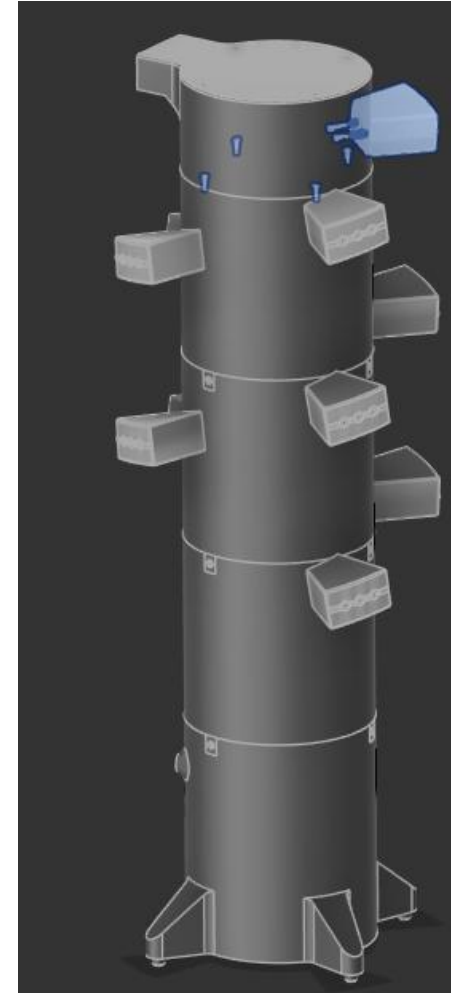
03-Assembly of the Helix frame KE-0401

8. Put the threaded M4 inserts into KE-0405 (3x in the flange that goes into KE-404). Bolt KE-0406 to KE-0405 from the inside using MC6S 4x12 (8x). Bolt KE-405 on to KE-404 using MC6S 4x12 (3x). Put the threaded M4 inserts in the top surface holes of KE-0405 (4x)



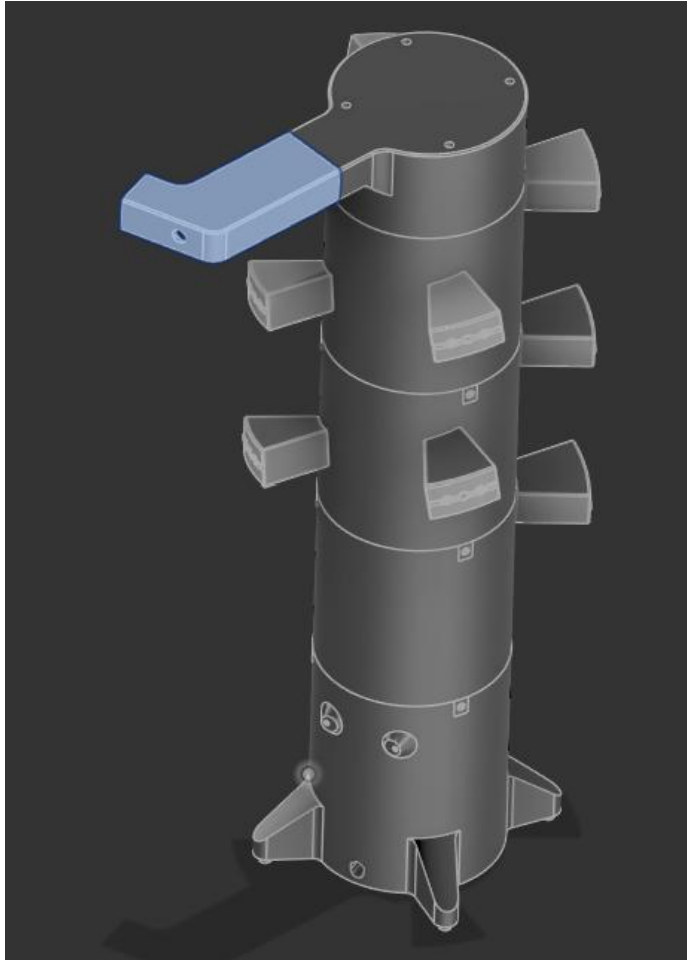
9. Bolt KE-0406 (1x) to KE-0407 using MC6S 4x40 (2x). Bolt KE-0407 to KE-0405 using MF6S 4x16 (4x).

Make sure to bolt KE-0406 to KE-0407 before KE-0407 is bolted to KE-0405.

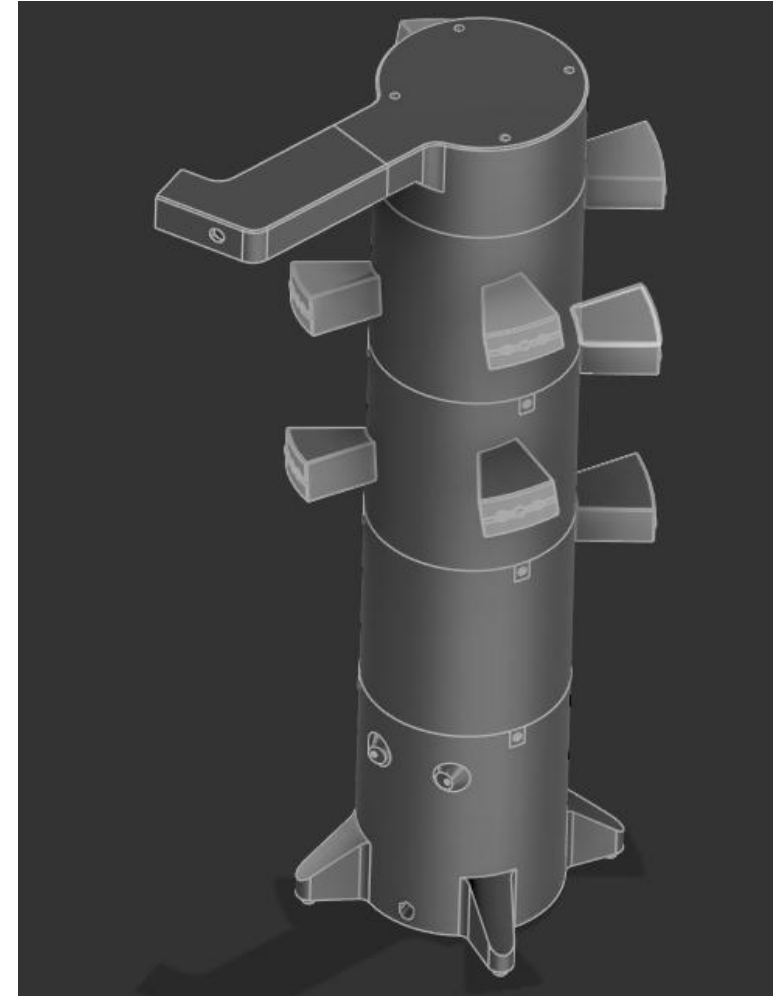


03-Assembly of the Helix frame KE-0401

5. Bolt KE-0418 on to KE-0407 using MC6S 4x12 (4x).



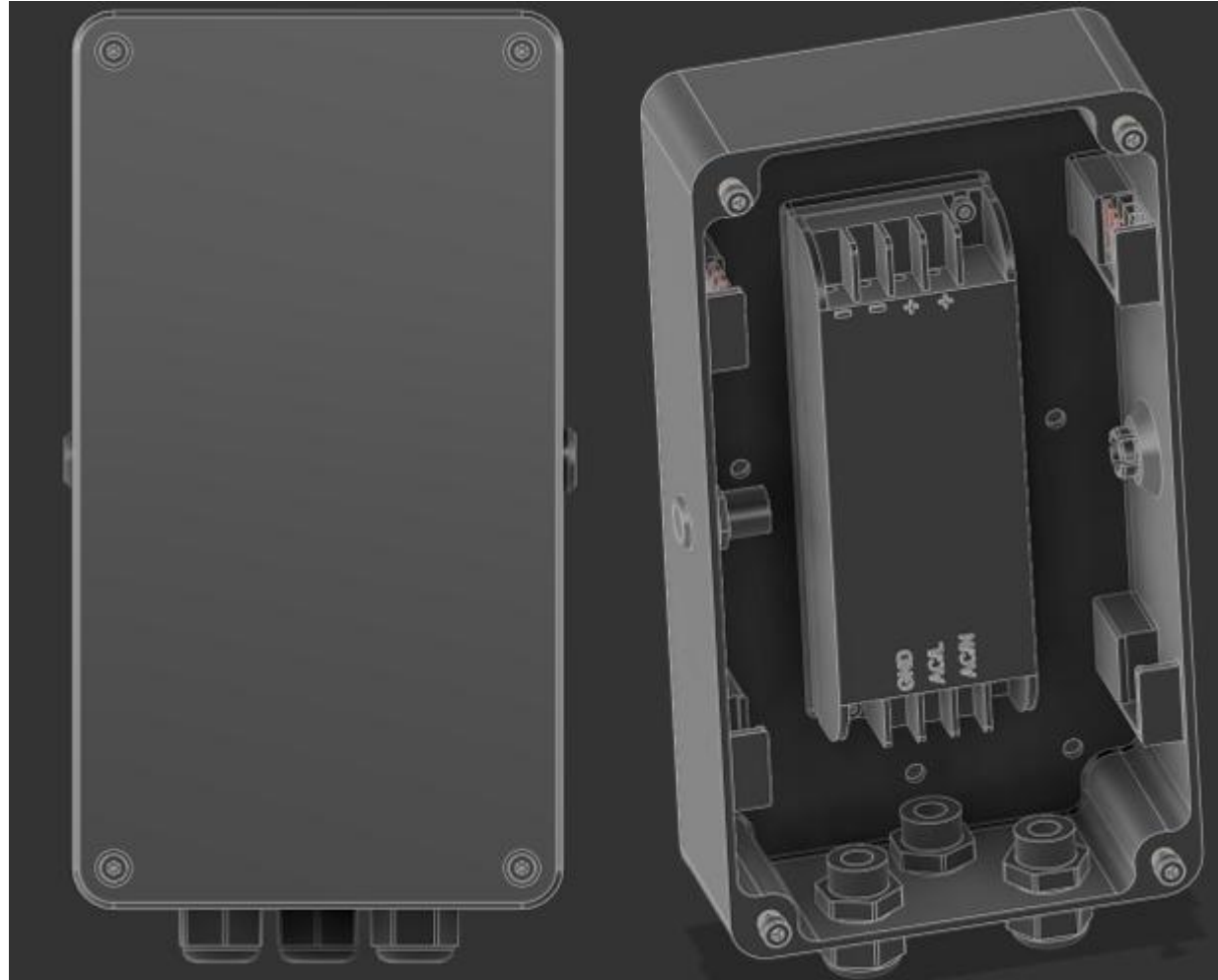
7. End result.



03-Electrical Cabinet KE-0175

5. This electrical cabinet has its own print profile and its own assembly manual. This electrical cabinet can be found here:

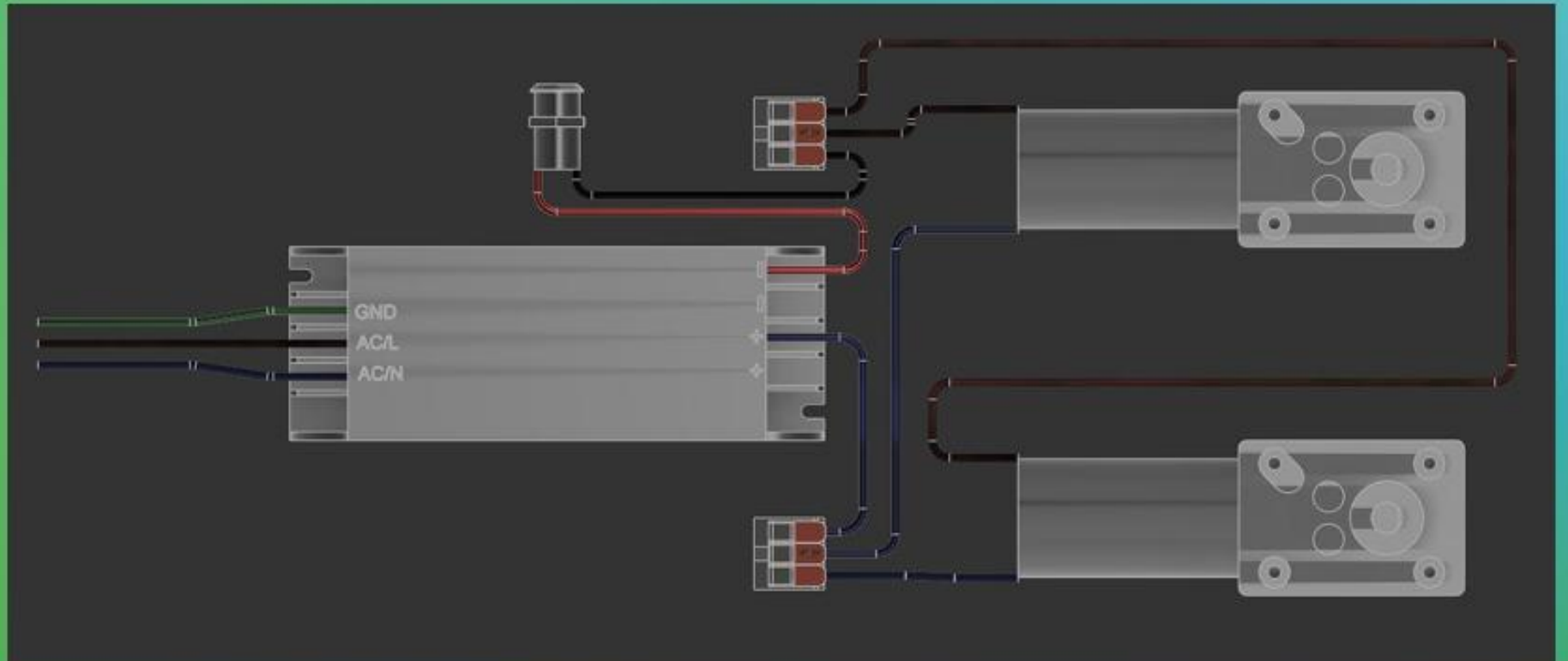
<https://makerworld.com/en/models/2052857-electrical-cabinet-for-110-220v-ac-to-12v-dc-ke-50#profileId-2215802>



03-Electrical Cabinet KE-0175

5. Use the schematics that will allow 2 motors to be controlled by one switch.

Schematics for one switch and 2 motors.

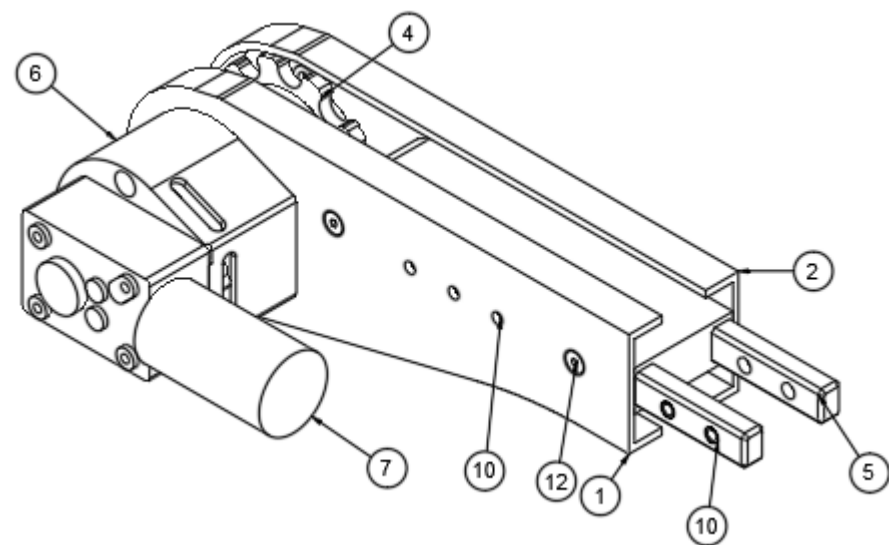


NOTE!

You will need a grounded power cord with whatever wall socket that fits the wall socket of your country. One end goes in the wall and the three cords in the other end goes into the transformer. From the transformer to the motors you need a 2-leader cable. I don't list these things here. Those are easy to buy in the hardware store of your country, which will guarantee that they fit the local electricity system.

03-3d-print and build the drive units KE-0002 & KE-0003

In this model two drive units will be used. The parts are exactly the same but the motor is placed on the left hand side for one of the drives and in the right hand side for the other. In the following step by step description one of the drive units will be described, but to build the other one, just turn the drive shaft the other way and fit the motor on the other side.

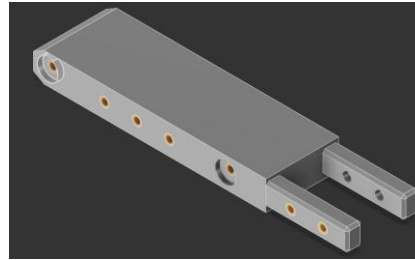


14	4	AB012	Threaded insert M3
13	4	ISO 4762 - M3 x 10 Steel 4.6 Plain	Hexagon Socket Head Cap ScrewISO 4762 - M3 x 10
12	7	DIN 7991 - M4x12 Steel 4.6 Plain	Hexagon Socket Countersunk Head Cap ScrewDIN 7991 - M4x12
11	4	DIN 7991 - M4x10 Steel 4.6 Plain	Hexagon Socket Countersunk Head Cap ScrewDIN 7991 - M4x10
10	22	AB016	Threaded insert M4
9	1	KE-0019	CLUTCH CONNECTOR
8	1	Flange clutch connector	Flange clutch connector
7	1	FTVOGUE480whpxgv7-06 (YVBL-WH)	DC-MOTOR 12V
6	1	KE-0018	MOTOR FLANGE
5	1	KE-0020	MOUNTING PLATE
4	1	KE-0015	DRIVE WHEEL
3	2	6804-2RS	BALL BEARING
2	1	KE-0017	GABLE, RIGHT
1	1	KE-0016	GABLE, LEFT
Item	Qty	Part Number	Description
Parts List			
Virmatt 2025.08.30			

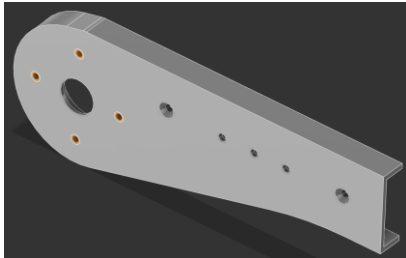
03-3d-print and build the drive units KE-0002 & KE-0003



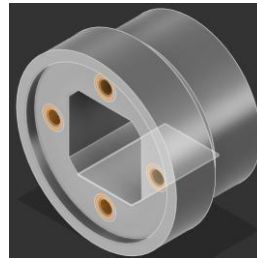
1. Print the parts on Plate 11 in the print profile KE-400 (Helix conveyor).3mf. Use PETG filament.



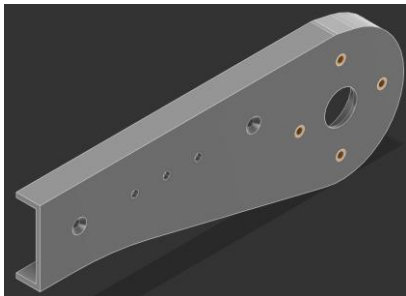
4. Put the M4 threaded inserts into KE-0020 (Mounting plate). 14pcs.



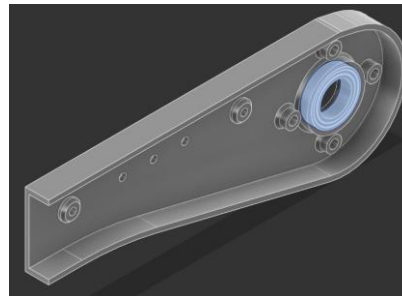
2. Put the M4 threaded inserts into KE-0016 (Gable Left). 4pcs.



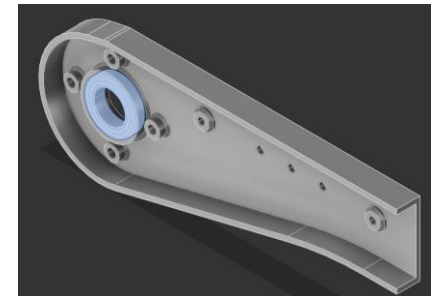
5. Put the M3 threaded inserts into KE-0019 (Clutch Connector). 4pcs.



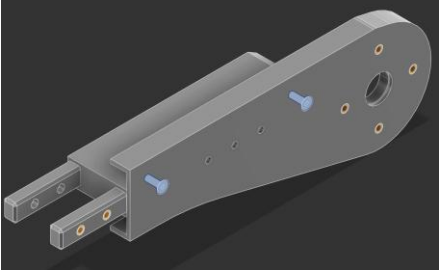
3. Put the M4 threaded inserts into KE-0017 (Gable Right). 4pcs.



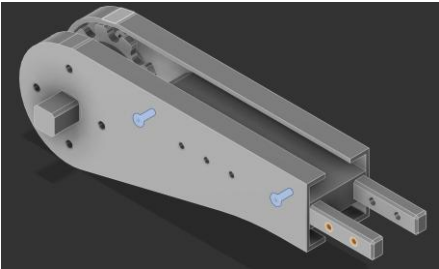
6. Put in the 6804-2RS1 ball bearings in the gables KE-0016 and KE-0017. 2pcs.



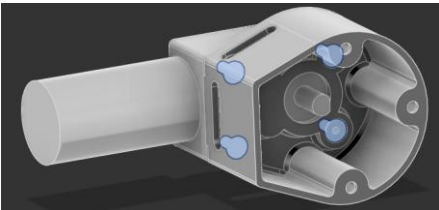
03-3d-print and build the drive unit KE-0002



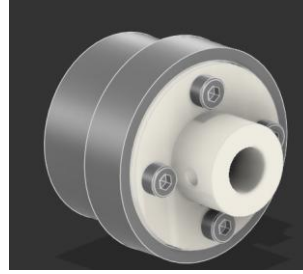
7. Bolt KE-0020 to KE-0017 using MF6S 4x16 screws. 2pcs.



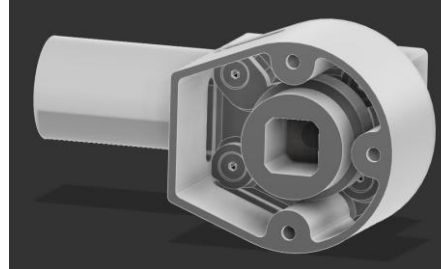
8. Put KE-0015 (Drive Wheel) Through the bearing of KE-0016 and KE-0017 and bolt KE-0016 to KE-0020 .



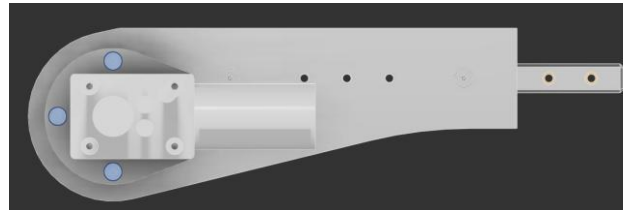
9. Bolt the gearbox to KE-0018 (Motor Flange) using MF6S 4x10. 4pcs.



10. Bolt the steel flange clutch to KE-0019 using MC6S 3x8. 4pcs.



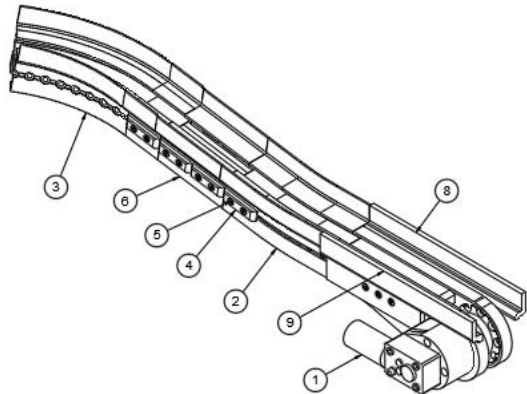
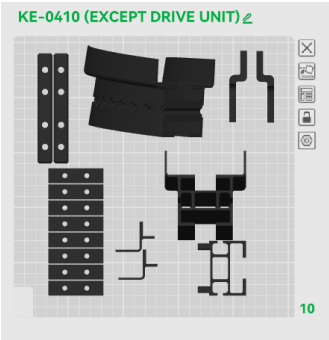
11. Bolt the steel flange to the drive shaft (let the drive shaft peek out ca 1mm). Tighten the set screw in the flange.



12. Bolt KE-0018 to whichever gable the drive shaft is located. Use MF6S 4x12. 3pcs. To build KE-0003, simply put the drive wheel with the drive shaft in the opposite direction and bolt the motor/flange on the opposite side (compared to this picture).

03-Assembly of the Helix lower end part KE-0410

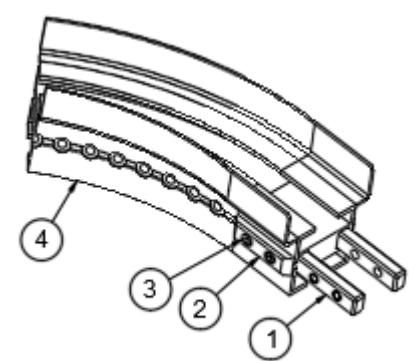
This is the lower part of the helix conveyor.
The material needed is shown in the BOM to the right. The drive unit KE-0003 has its own assembly instruction.
Print the parts on plate 10. Follow the step by step instruction starting on the next page.



9	1	KE-0421	GUIDE RAIL FOR DRIVE UNIT, RIGHT
8	1	KE-0420	GUIDE RAIL FOR DRIVE UNIT, LEFT
7	2	KE-0050	CONNECTING ROD
6	1	KE-0413	CHAIN INSERTION SECTION
5	18	ISO 4762 - M4 x 12 Steel 4.6 Plain	Hexagon Socket Head Cap Screw ISO 4762 - M4 x 12
4	6	KE-0051	CLAMP, SHORT
3	1	KE-0210	LOWER HELIX END PART
2	1	KE-0408	VERTICAL BEND 9 DEGREES UP
1	1	KE-0003	DRIVE UNIT
Item	Qty	Part Number	Description
Parts List			

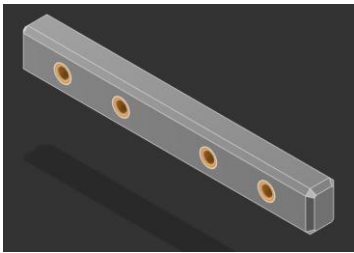
03-Assembly of the Helix lower end section part KE-0210

1. Begin with the end section KE-0210. This is a subassembly to KE-0410.

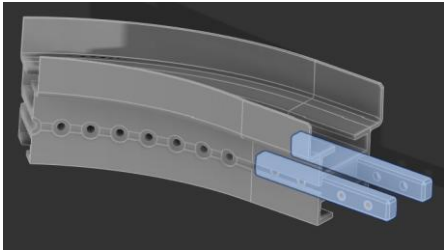


4	1	KE-0422	HELIX SECTION, LOWER END
3	4	ISO 4782 - M4 x 12 Steel 4.8 Plain	Hexagon Socket Head Cap Screw ISO 4782 - M4 x 12
2	2	KE-0051	CLAMP, SHORT
1	2	KE-0050	CONNECTING ROD
Item	Qty	Part Number	Description

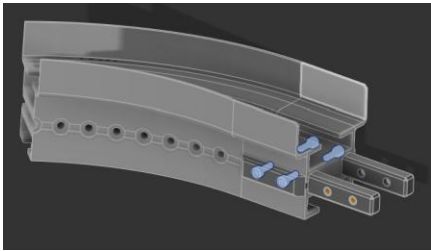
2. Put the threaded M4 inserts into KE-0050.



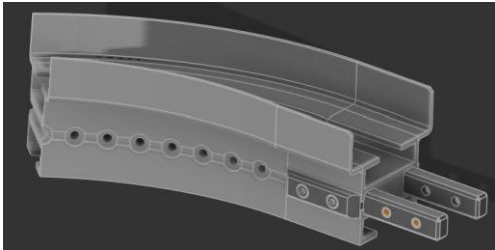
3. Push the KE-0050's all the way in to the t-slots of KE-0422.



4. Bolt the clamps, KE-0051 like in the picture, using MC6S 4X12 screws (4x).

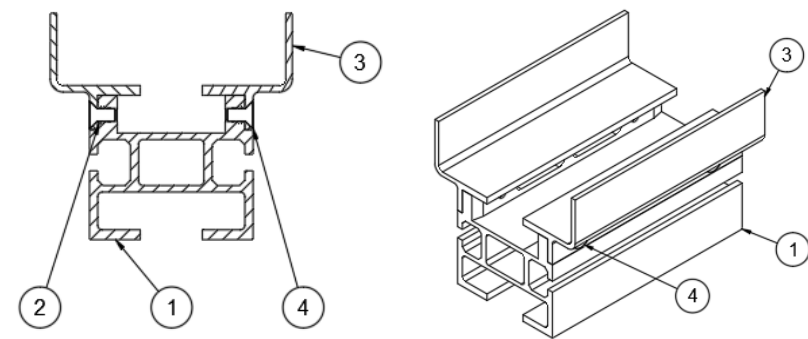


5. End result KE-0210.



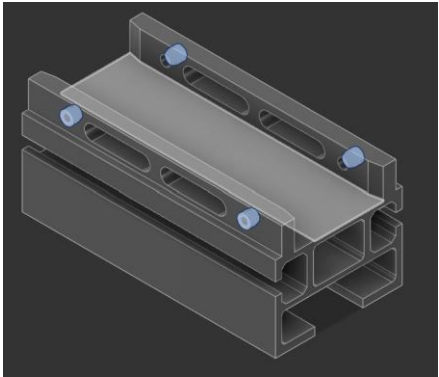
03-Assembly of the Helix chain insertion section KE-0413

1. This is a chain insertion section. This is a subassembly to KE-0410.

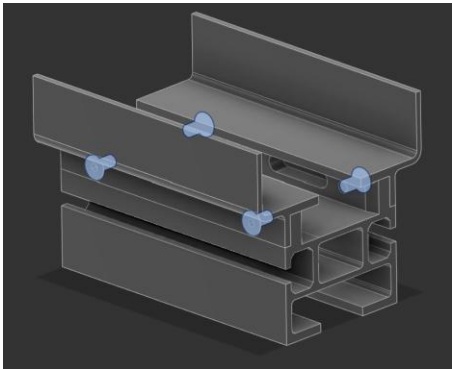


4	4	DIN EN ISO 10642 - M3 x 8 Steel 8.8 Plain	Hexagon Socket Countersunk Head Screw DIN EN ISO 10642 - M3 x 8
3	2	KE-0414	GUIDE
2	4	AB012	Threaded insert M3
1	1	KE-0045	BEAM
Item	Qty	Part Number	Description
Parts List			

2. Put the threaded M3 inserts (4x) into KE-0045.

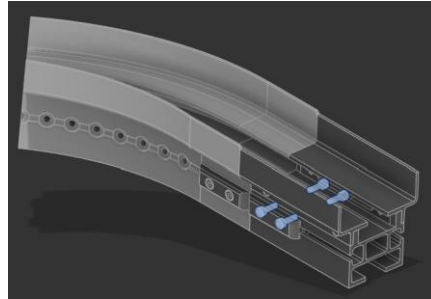


3. Bolt on the guides KE-0414 (2x) using MF6S 3x8 (4x).

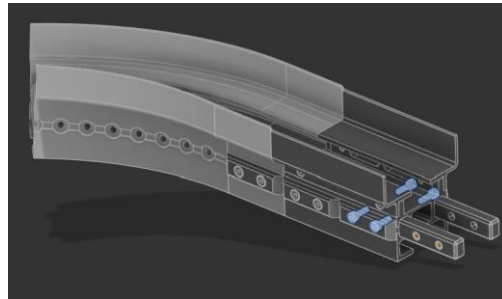


03-Final assembly of KE-0410

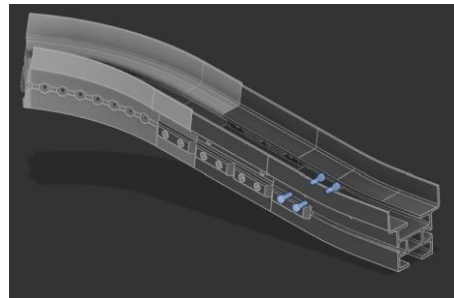
1. Fit KE-0210 and KE-0413 together using clamps KE-0051 (2x) and screws MC6S 4x12 (4x)



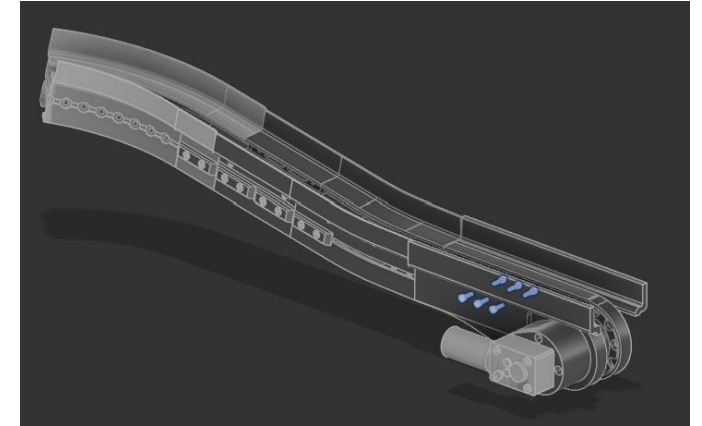
2. Fit the connecting rods KE-0050 (2x) using clamps KE-0051 (2x) and screws MC6S 4x12 (4x).



3. Fit the vertical bend KE-0408 using clamp and screws just like the previous step.

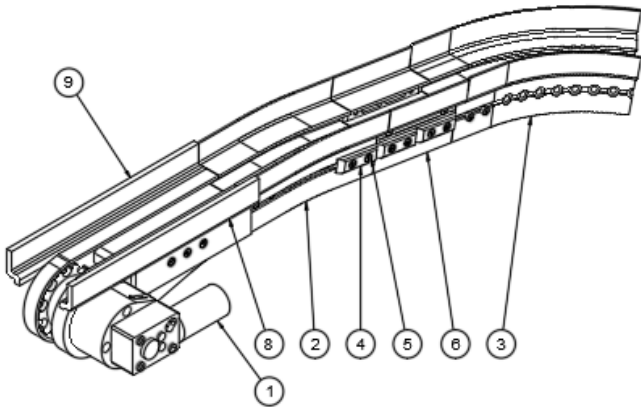
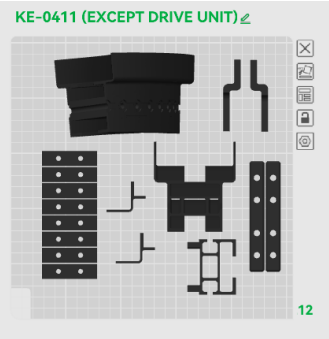


4. Fit the drive unit KE-0003, the guide KE-0420 and KE-00421. Fit the guides with MC6S 4x12 (6x). Note that the drive unit is not clamped to the rest of the assembly just yet. Later on, the leg support will be bolted to the connecting rods of the drive unit.



03-Assembly of the Helix upper end part KE-0411

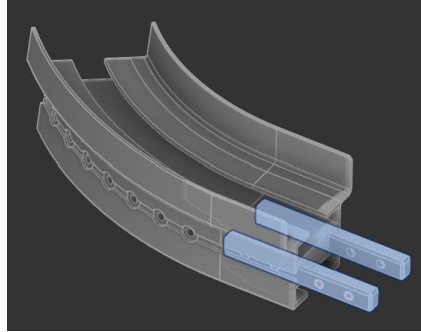
This is the upper part of the helix conveyor.
The material needed is shown in the BOM to the right. The drive unit KE-0002 has its own assembly instruction.
Print the parts on plate 12. Follow the step by step instruction starting on the next page.



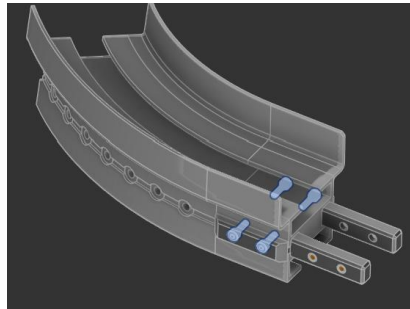
9	1	KE-0421	GUIDE RAIL FOR DRIVE UNIT, RIGHT
8	1	KE-0420	GUIDE RAIL FOR DRIVE UNIT, LEFT
7	2	KE-0050	CONNECTING ROD
6	1	KE-0413	CHAIN INSERTION SECTION
5	20	ISO 4762 - M4 x 12 Steel 4.6 Plain	Hexagon Socket Head Cap Screw ISO 4762 - M4 x 12
4	7	KE-0051	CLAMP, SHORT
3	1	KE-0212	HELIX UPPER END PART
2	1	KE-0409	VERTICAL BEND 9 DEGREES DOWN
1	1	KE-0002	DRIVE UNIT
Item	Qty	Part Number	Description
Parts List			

03-Assembly of the Helix upper end part KE-0212

1. Begin with the end section KE-0212. This is a subassembly to KE-0411. Put the connecting rods KE-0050 into KE-0423 all the way in. Make sure to put in the M4 inserts into the rods as described earlier.

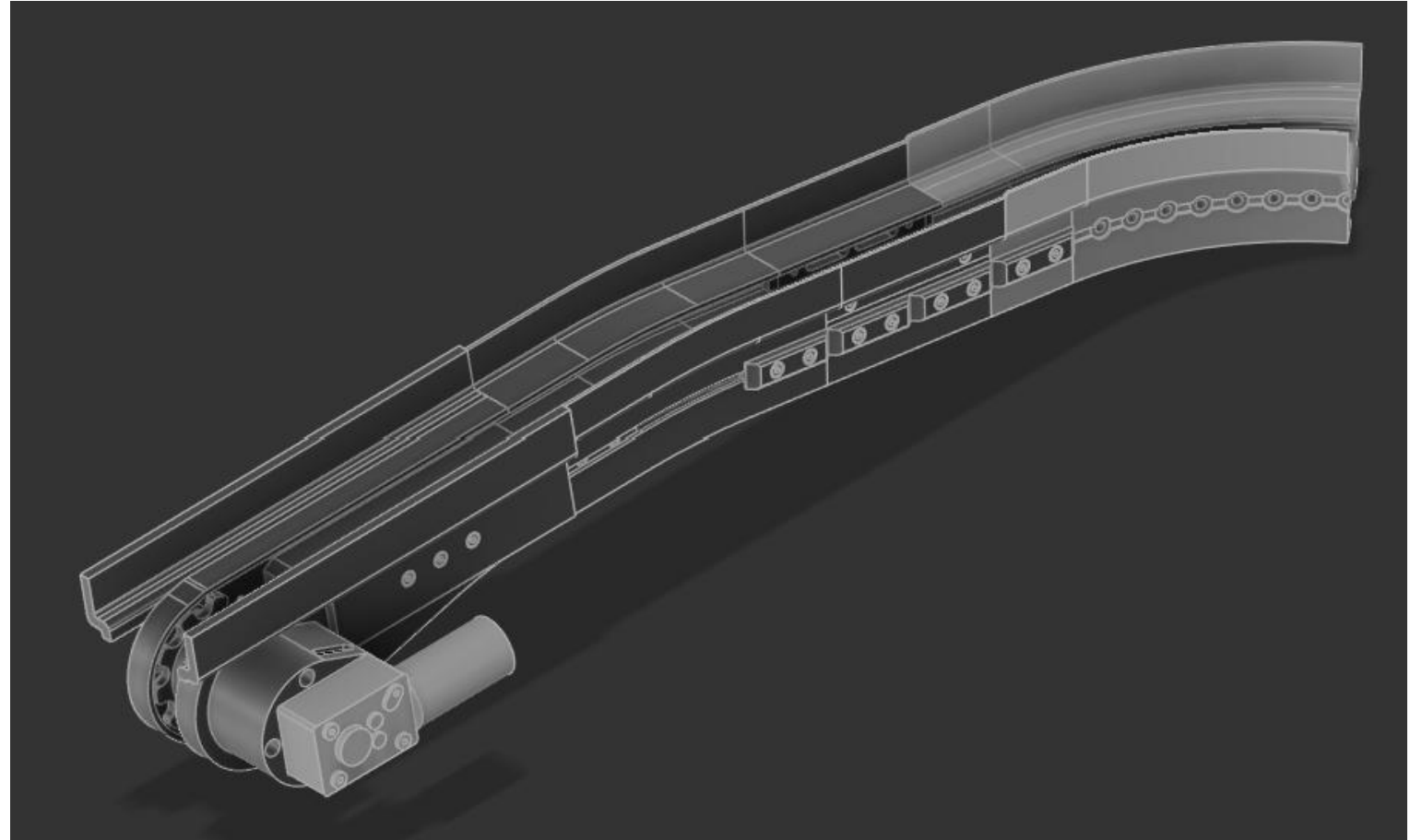


2. Clamp the rods to the beam section using KE-0050 (2x) and MC6S 4x12 (4x).



03-Final assembly of KE-0412

1. Put all the parts together in the same way as on KE-0410 described earlier. The only difference is that now the vertical bend is KE-0409 (bending downwards) and the drive unit is KE-0002 (motor on the left side).



03-Assembly of Conveyor Chain KE-0149

- Print the parts on plate 15,16 & 17 of the print profile KE-400 (Helix conveyor).3mf.
- NOTE! The material is very important! **Use Bambulab PETG HF for the chain body.** Not all PETG filaments work for the chain! The friction cleat is made out of Bambulab TPU 95 HF.
- You will need **418 links** for this conveyor.



1. You need two chain bodies KE-0144, one hinge KE-0058, two friction cleats KE-0145 and one pin 3x30mm.



2. Put the hinge in the chain body



3. Drill through both chain bodies and the hinge with a 3mm drill.



4. Push the pin through it all. Use your thumb or some tool.



5. Put some glue and push the TPU friction cleat into the chain body



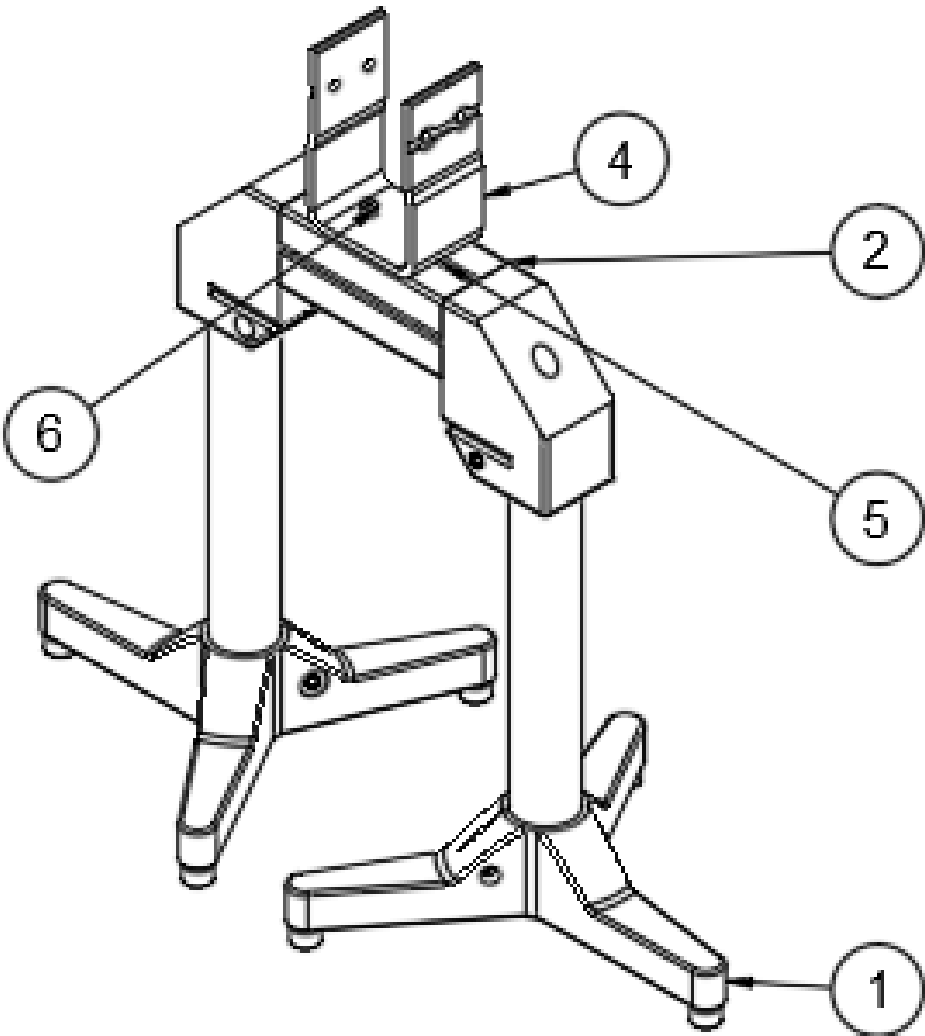
6. End result

03-3d-print and build KE-0417



1. Print the parts on 14. Use PETG filament.
2. Find the drawing further down in the documentation and use it to assemble this leg support. It's pretty straight forward and does not need a more thorough instruction than the drawing itself.

6	4	ISO 4762 - M4 x 12 Steel 4.6 Plain	Hexagon Socket Head Cap Screw ISO 4762 - M4 x 12
5	2	KE-0053	SLOT NUT
4	1	KE-0119	CONVEYOR BRACKET
3	2	AB016	Threaded insert M4
2	1	KE-0114	SUPPORT BEAM, 94MM
1	2	KE-0415	LEG SUPPORT FOR WEDGE CONVEYOR, KE 50
Item	Qty	Part Number	Description
Parts List			

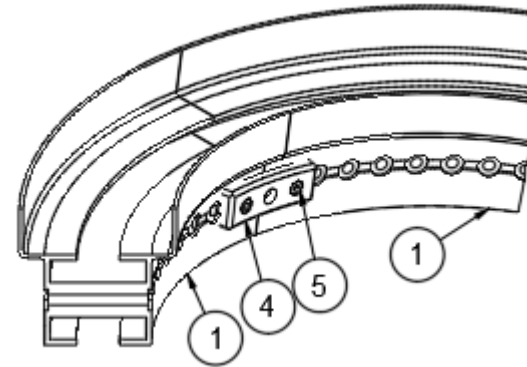


03-3d-print and build KE-0419



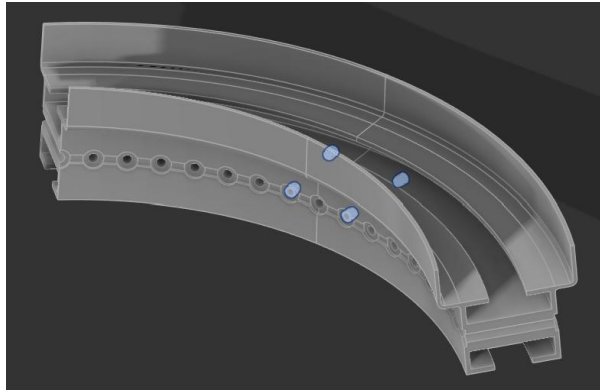
1. Print plate 09 and 13.

5	4	ISO 4762 - M4 x 12 Steel	Hexagon Socket Head Cap
		4.8 Plain	Screw ISO 4762 - M4 x 12
4	1	KE-0215	CONNECTING STRIP, INNER
3	1	KE-0213	CONNECTING STRIP
2	4	AB016	Threaded insert M4
1	2	KE-0211	45 DEGREE HELIX SEGMENT
Item	Qty	Part Number	Description
Parts List			

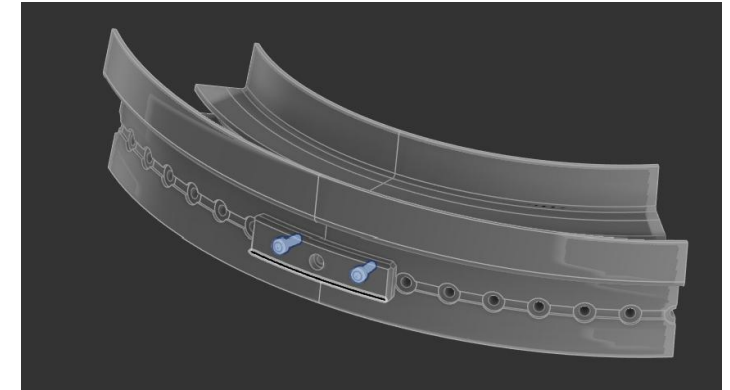


03-3d-print and build KE-0419

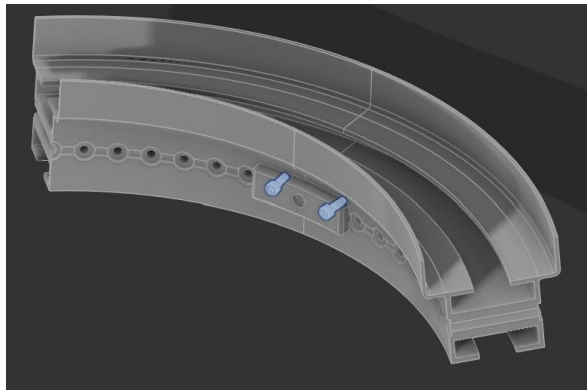
1. Put the M4 threaded inserts in KE-0211 (4x).
In the picture 2x KE-0211 is shown, each with 2x of M4 inserts.



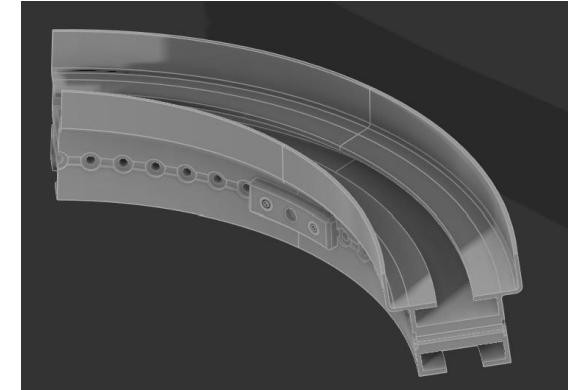
3. Bolt the connecting strip KE-0213 to the both conveyor beam segments, using MC6s 4x12 (2x).



2. Bolt the connecting strip KE-0215 to the both conveyor beam segments, using MC6S4x12 (2x).



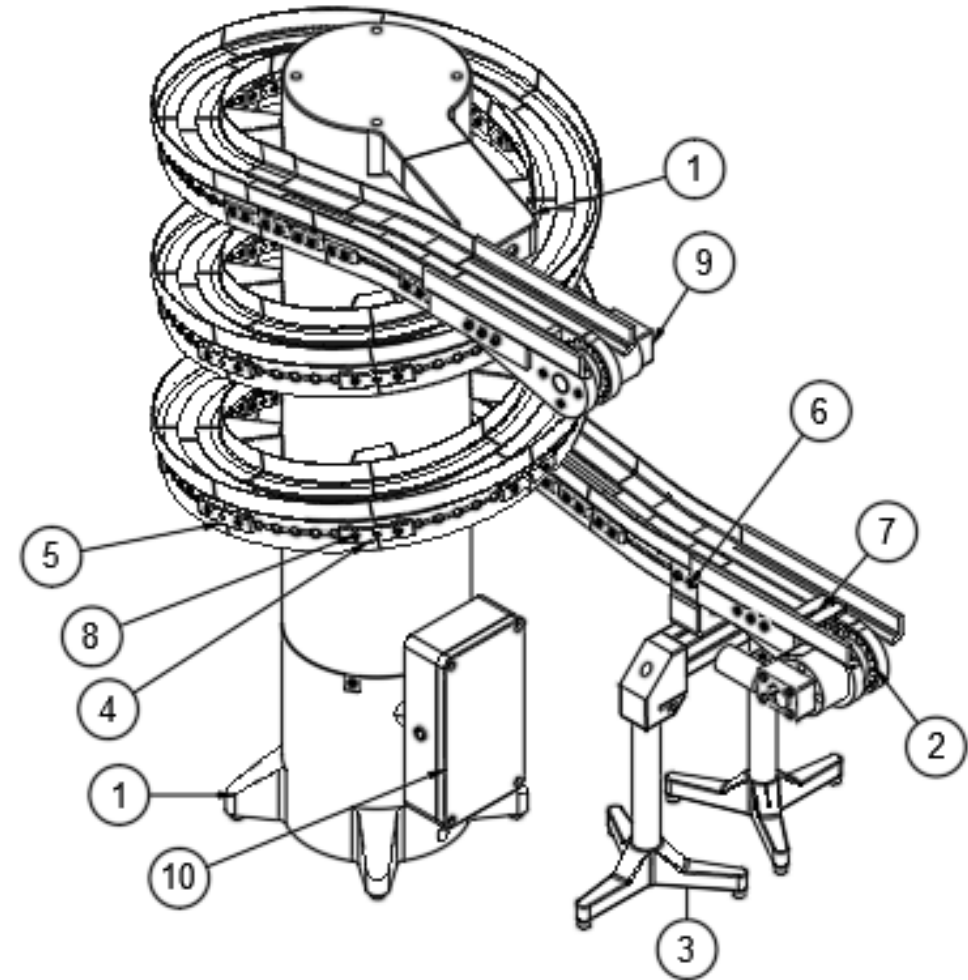
3. End result



03-3d-print and build KE-0400. Final assembly.

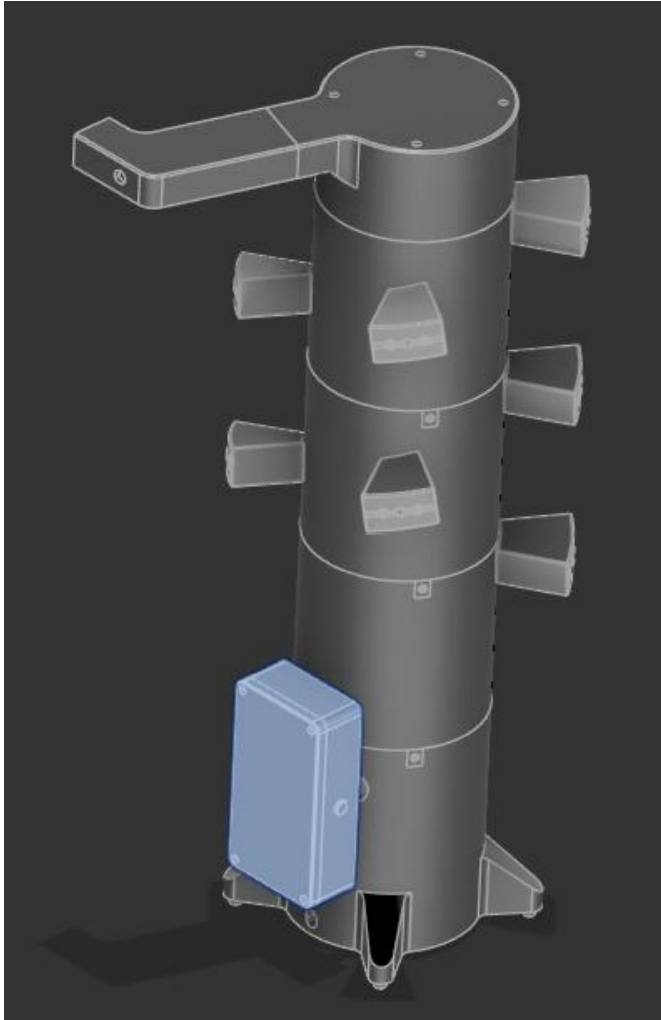
1. Final assembly.

10	1	KE-0175	ELECTRICAL CABINET
9	1	KE-0411	CONVEYOR, UPPER END SECTION
8	20	MC6S 4x60	Hexagon Socket Head Cap Screw ISO 4762
7	1	KE-0149	FRICTION CHAIN, KE 50
6	8	ISO 4762 - M4 x 12 Steel 4.6 Plain	Hexagon Socket Head Cap Screw ISO 4762 - M4 x 12
5	9	KE-0419	90 DEGREE HELIX SEGMENT
4	11	KE-0213	CONNECTING STRIP
3	1	KE-0417	SUPPORT FRAME
2	1	KE-0410	CONVEYOR, LOWER END SECTION
1	1	KE-0401	FRAME FOR HELIX, Helix conveyor KE 50
Item	Qty	Part Number	Description
Parts List			

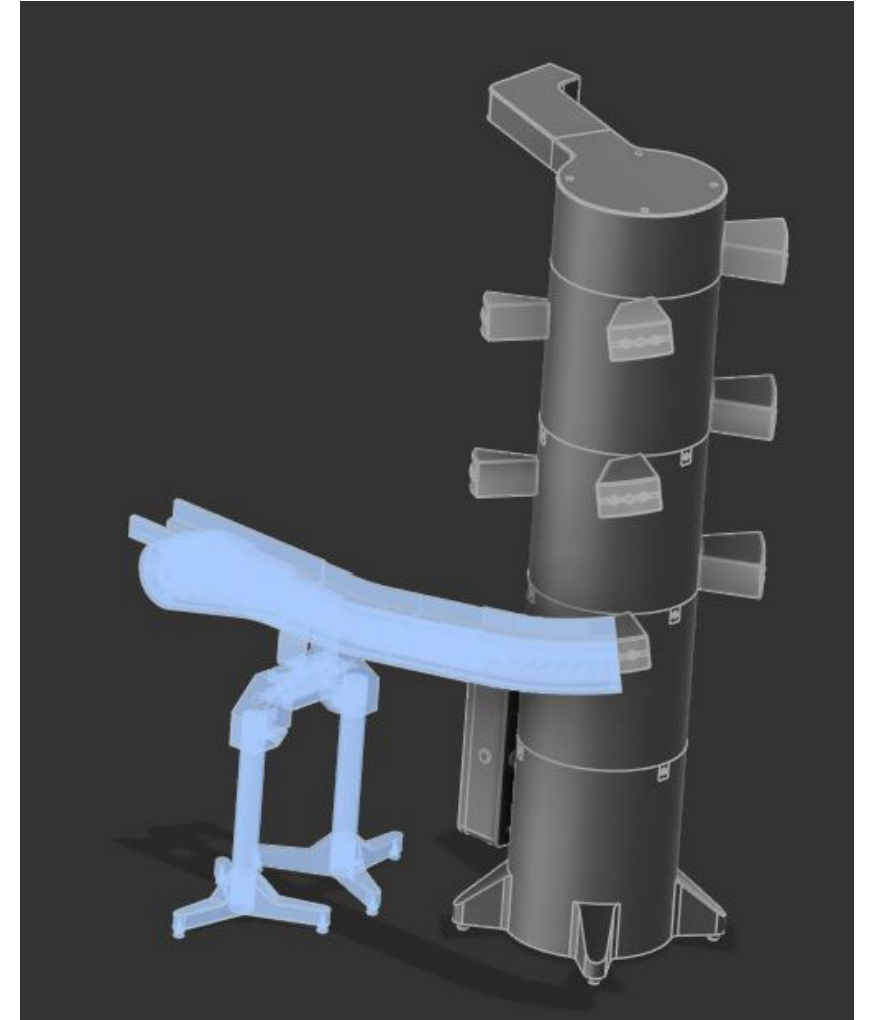


03-3d-print and build KE-0400. Final assembly.

1. Put the electrical cabinet KE-0175 on the frame KE-0401 using MC6S 4x12 (2x). You need to remove the lid from the cabinet in order to bolt it to the frame.

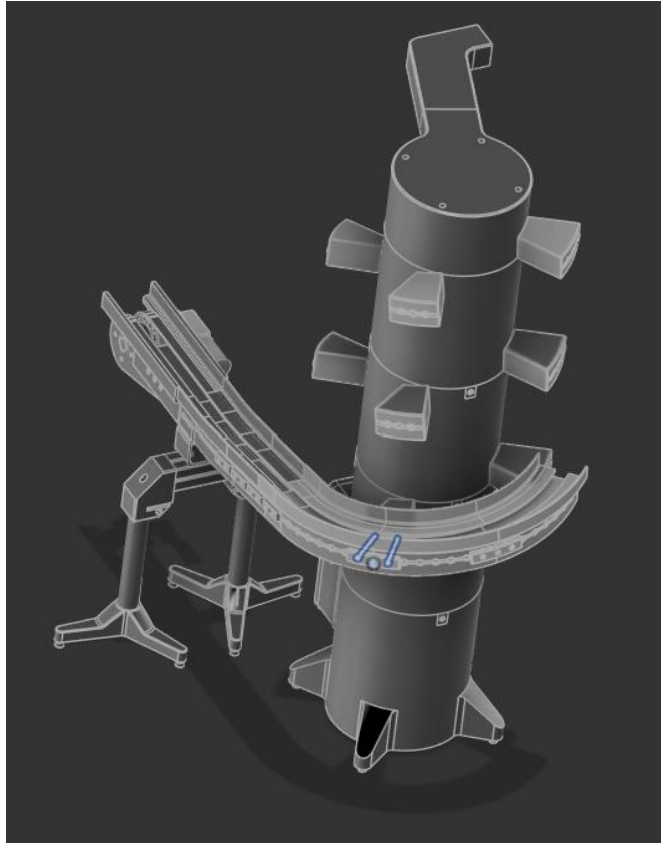


2. Bolt the leg support KE-0417 to the conveyor section KE-0410 using MC6S 4x12 (4x).

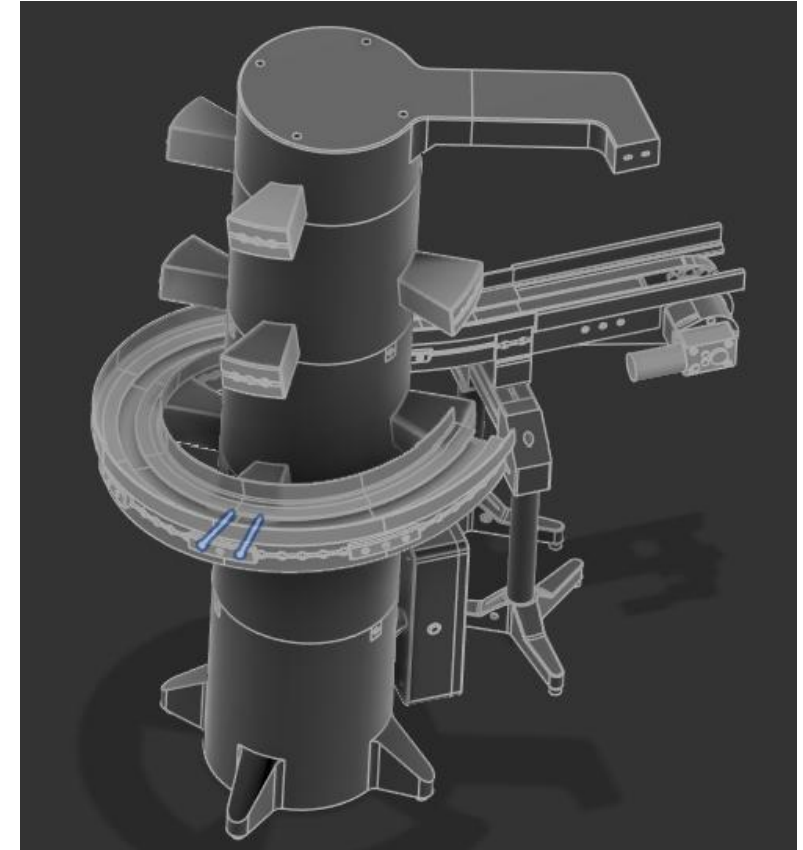


03-3d-print and build KE-0400. Final assembly.

3. Bolt KE-0210 and KE-0419 to the frame KE-0401 via a connecting strip (KE-0213). Use MC6S 4x60 (2x)



4. Bolt the next 45 degree helix section (KE-0419) to the frame KE-0401 using MC6S 4x60 (2x). Continue repeating this procedure until the last conveyor bracket (KE-0406) is reached.

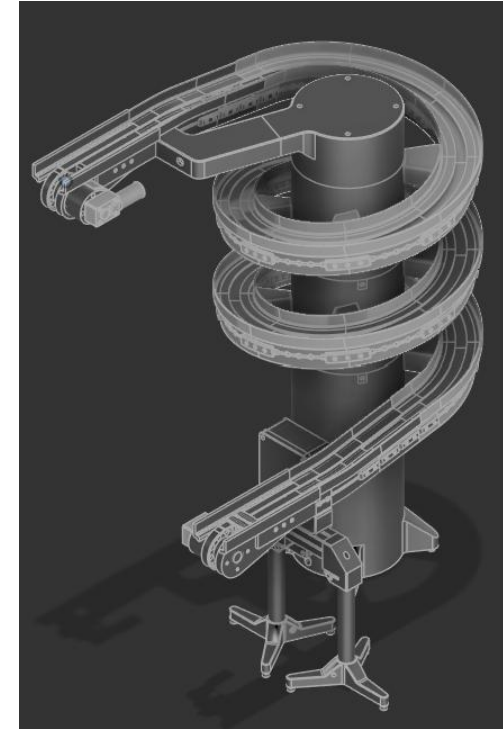


03-3d-print and build KE-0400. Final assembly.

5. Bolt the conveyor section KE-0411 to the frame using a connecting strip KE-0213 and MC6S 4x60 (2x). On the other side of the conveyor segment, use MC6S 4x12 (2x) as in the picture.



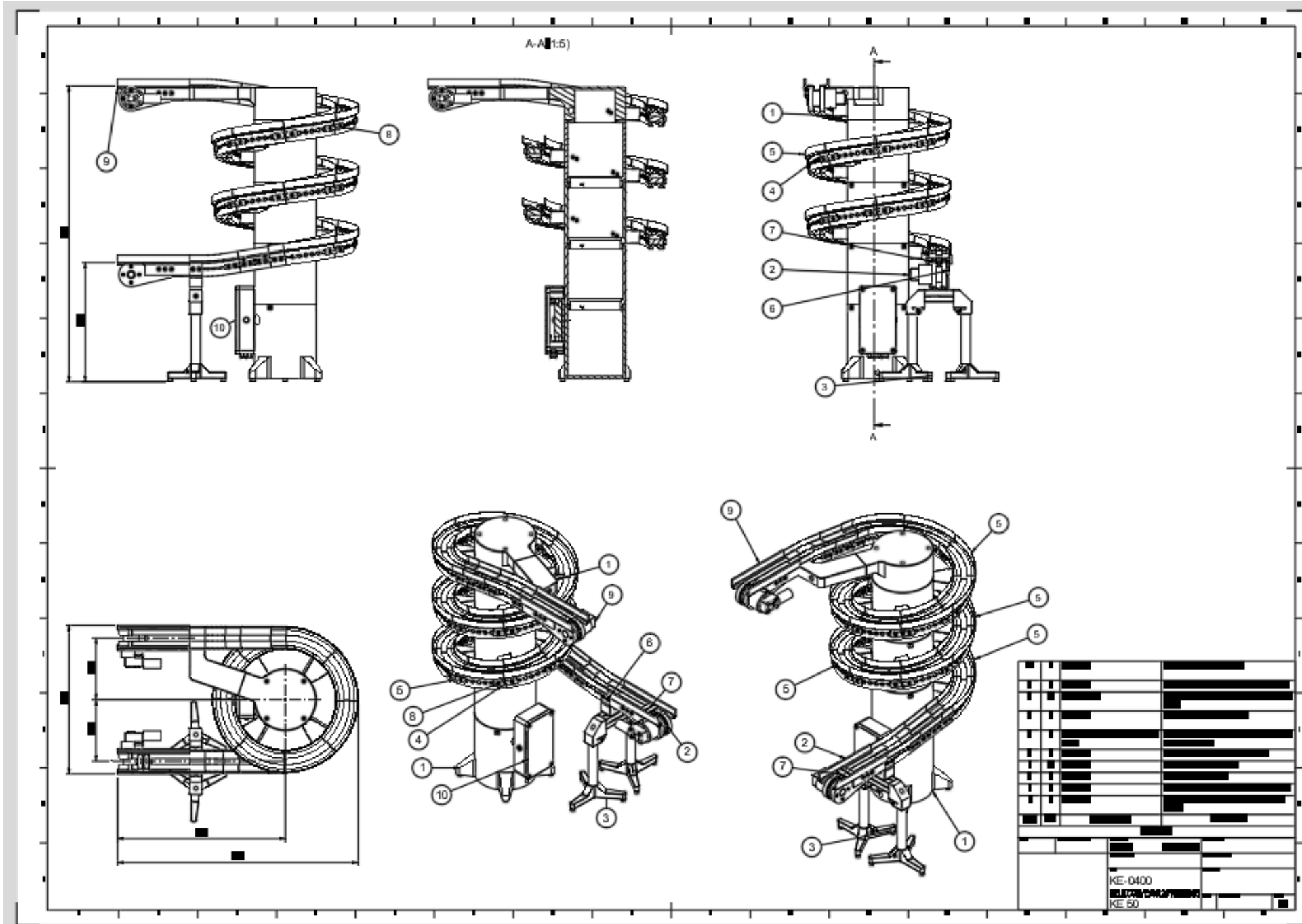
6. Use the wiring schematics for one switch and 2 motors described in the separate assembly manual for KE-0175. Make sure that the motors run in a direction that makes them help each other. To change rotational direction of the motors, just reverse the poles.



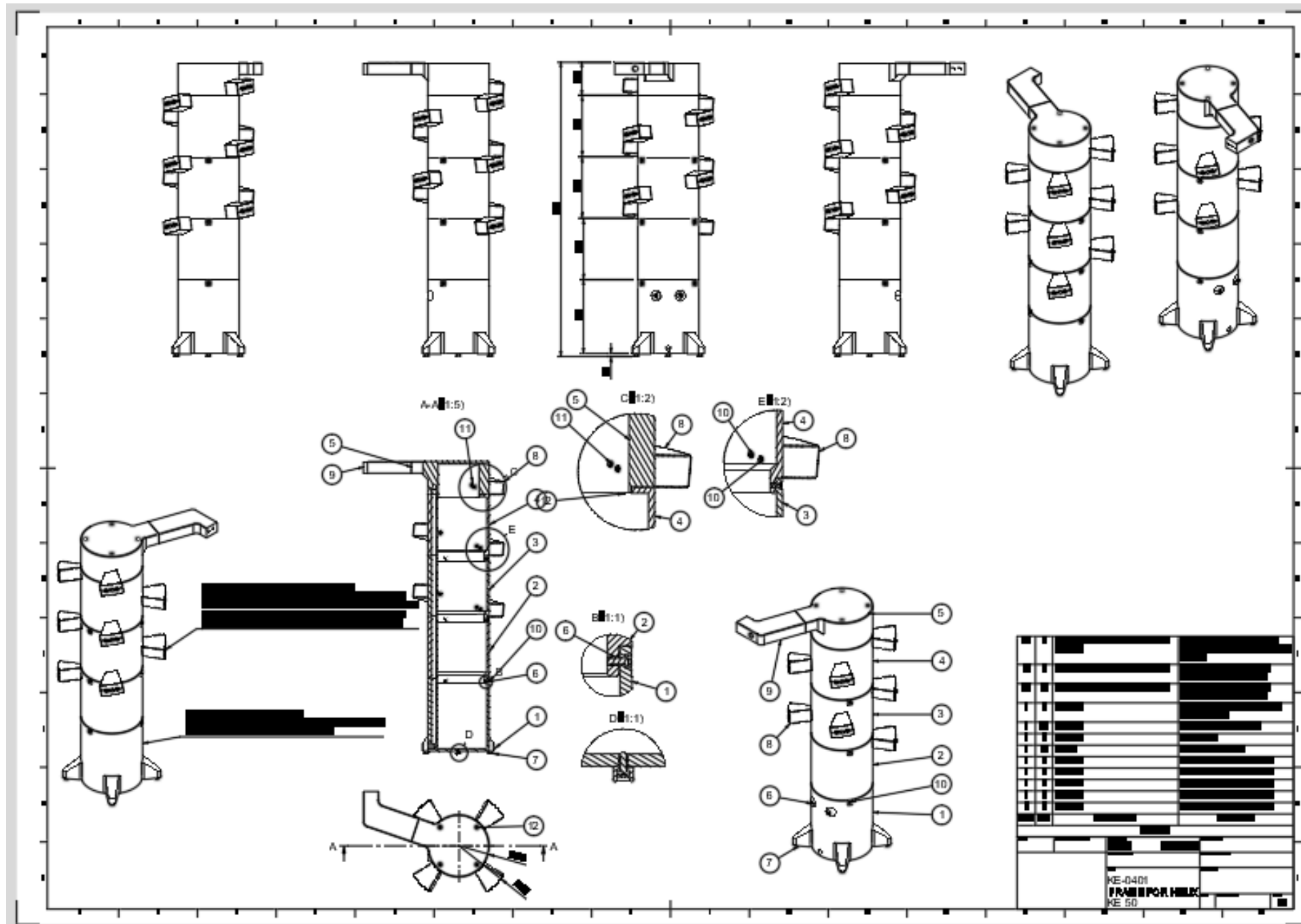
03- Chain installation and test run and tips

- This is best described in a video. Please follow this Youtube link to see how it is done:
- Chain assembly instruction: https://youtu.be/MUYROsvMsmA?si=Foog_Xuqwu9ZsW9d
- Chain installation instruction (different chain type but same procedure): <https://youtu.be/QmK309wBs9c>
- Test run 33cl cans: <https://youtu.be/I0arsVEZ36o>
- Just a walk around the conveyor: https://youtube.com/shorts/QzKh2leah_A

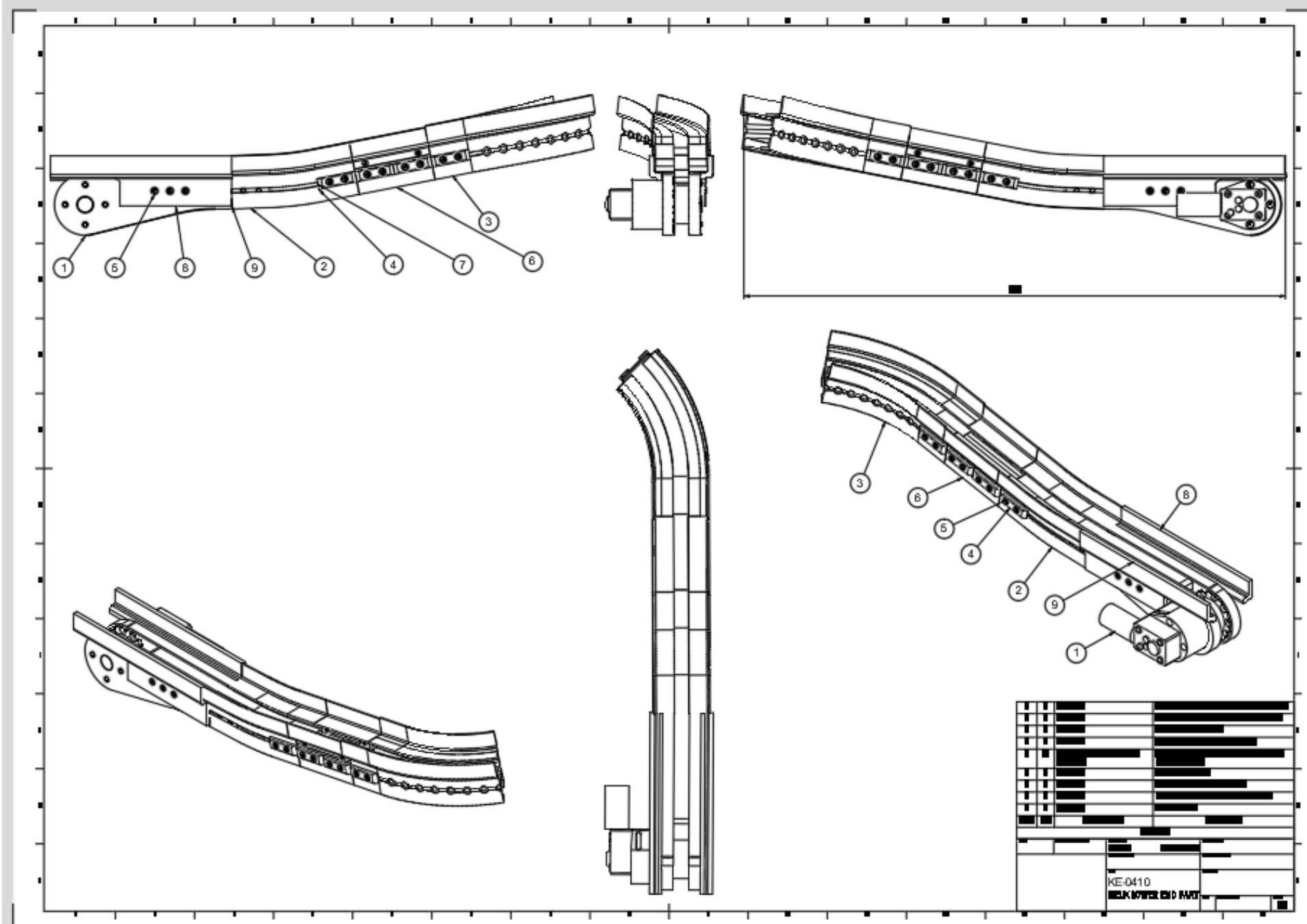
04-Drawings



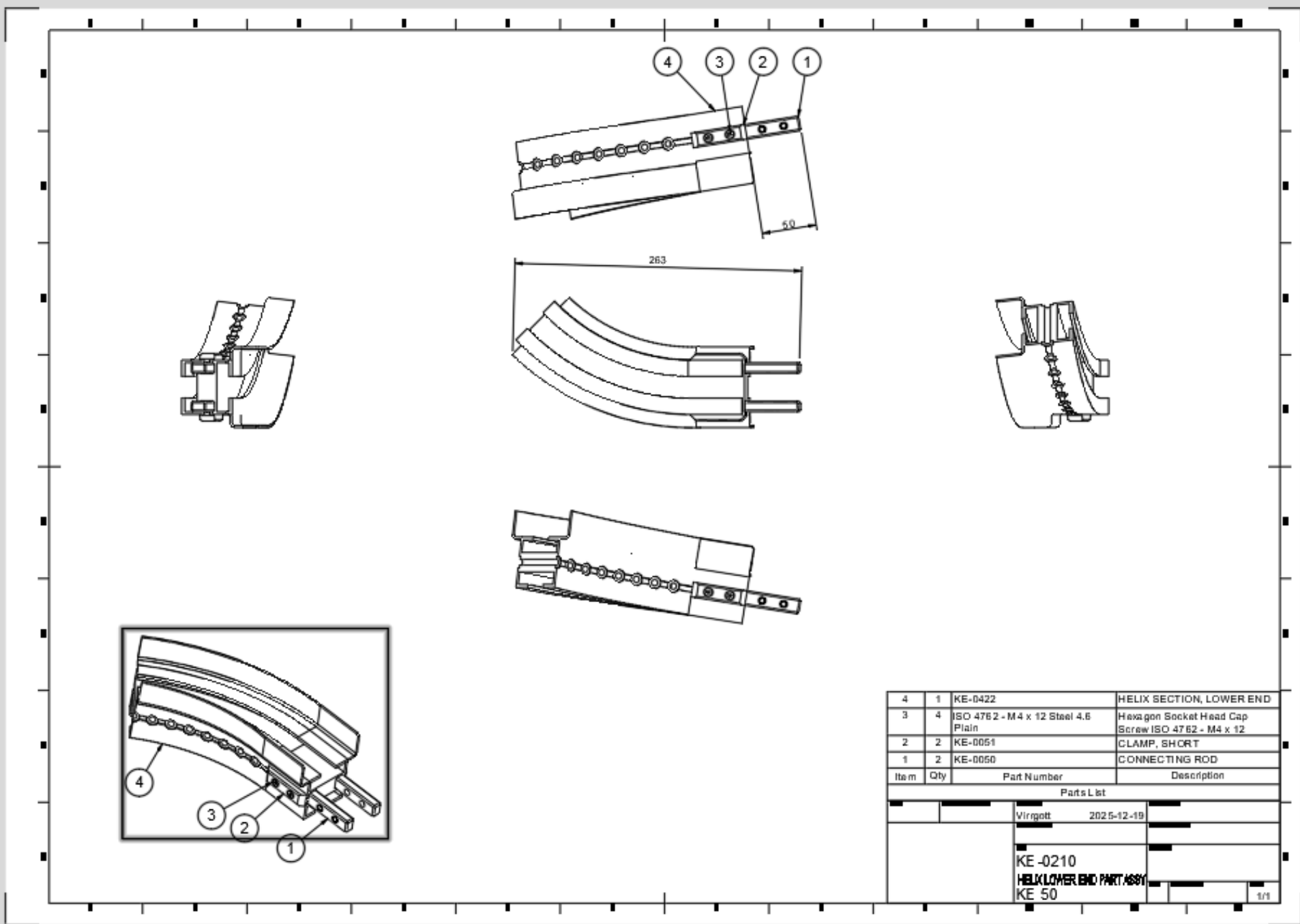
04-Drawings



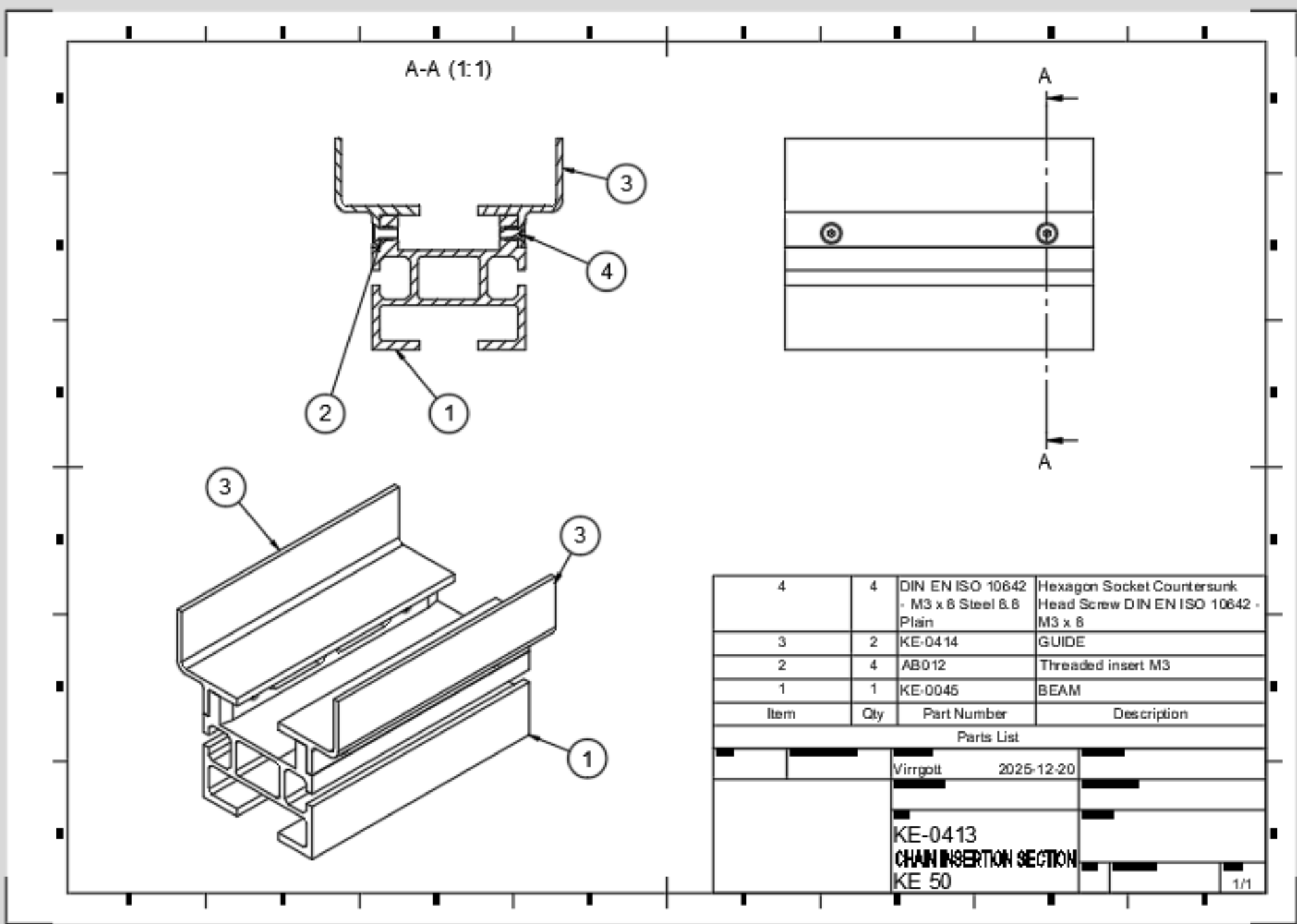
04-Drawings



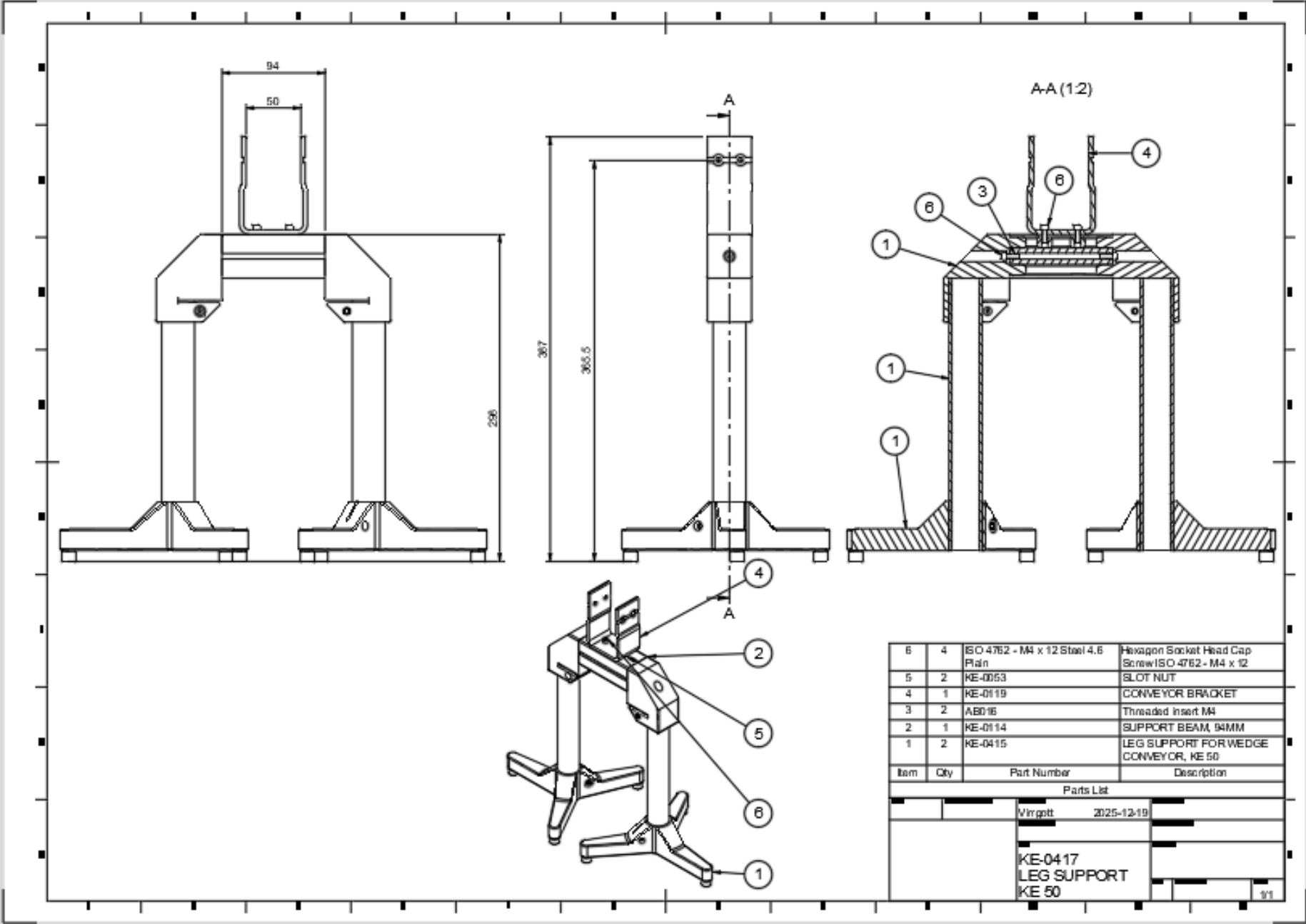
04-Drawings



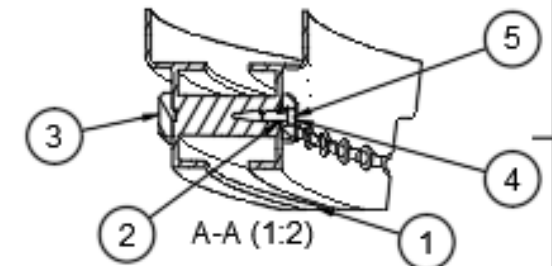
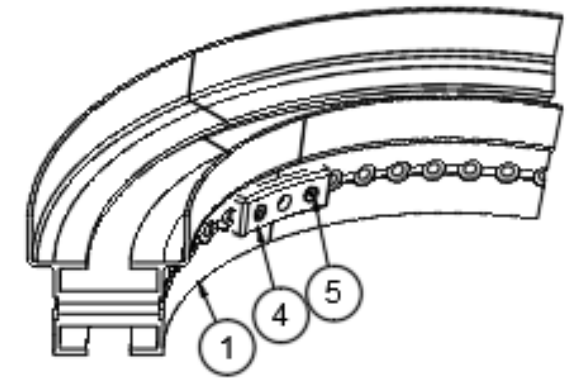
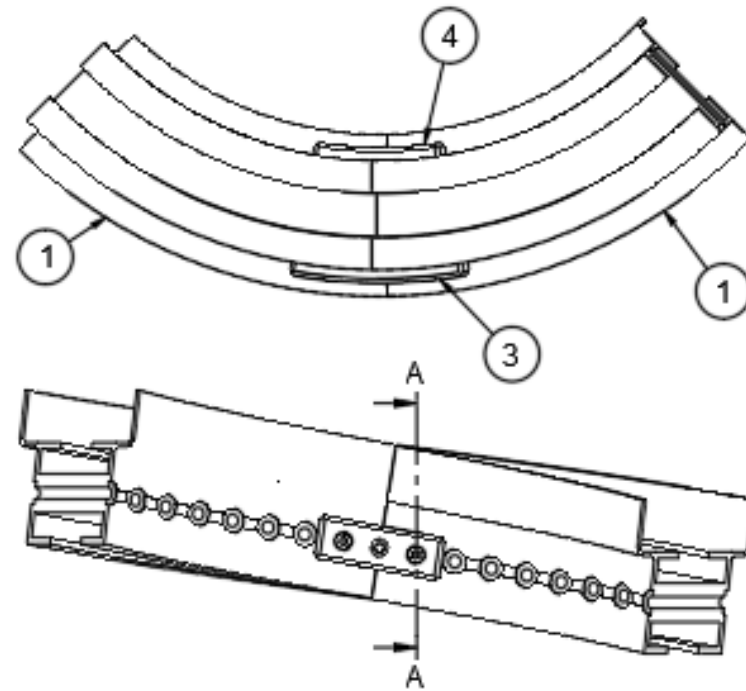
04-Drawings



04-Drawings



04-Drawings

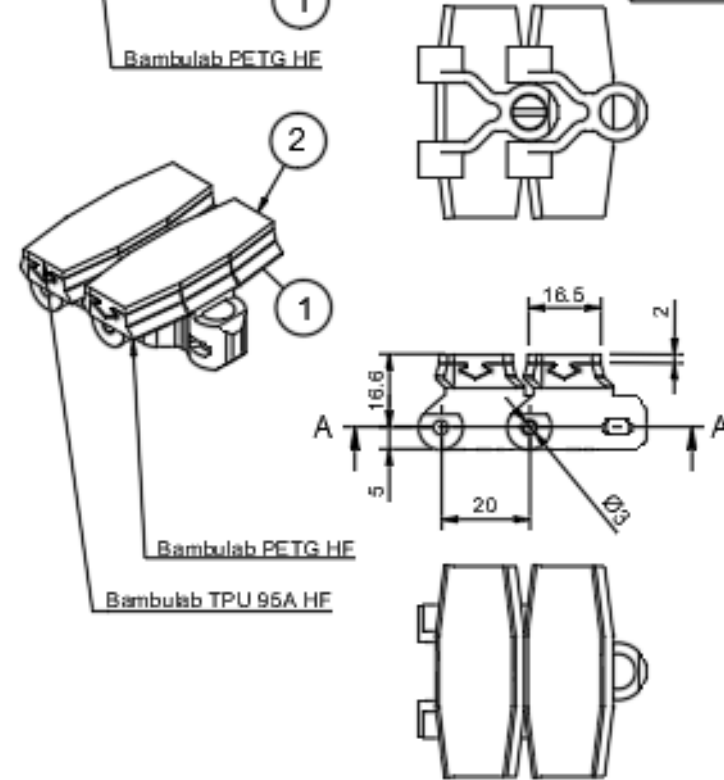
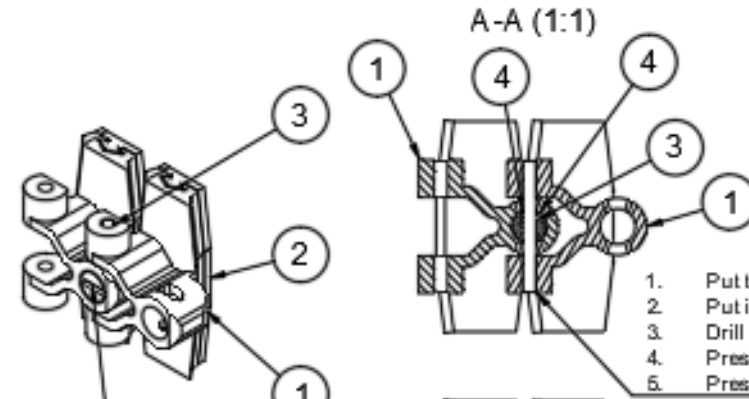


5	4	ISO 4762 - M4 x 12 Steel	Hexagon Socket Head Cap
		4.8 Plain	Screw ISO 4762 - M4 x 12
4	1	KE-0215	CONNECTING STRIP, INNER
3	1	KE-0213	CONNECTING STRIP
2	4	AB016	Threaded insert M4
1	2	KE-0211	45 DEGREE HELIX SEGMENT
Item	Qty	Part Number	Description

Parts List

Virgott	2025-12-19	
KE-0419		
90 DEGREE HELIX SEGMENT		
KE 50		1/1

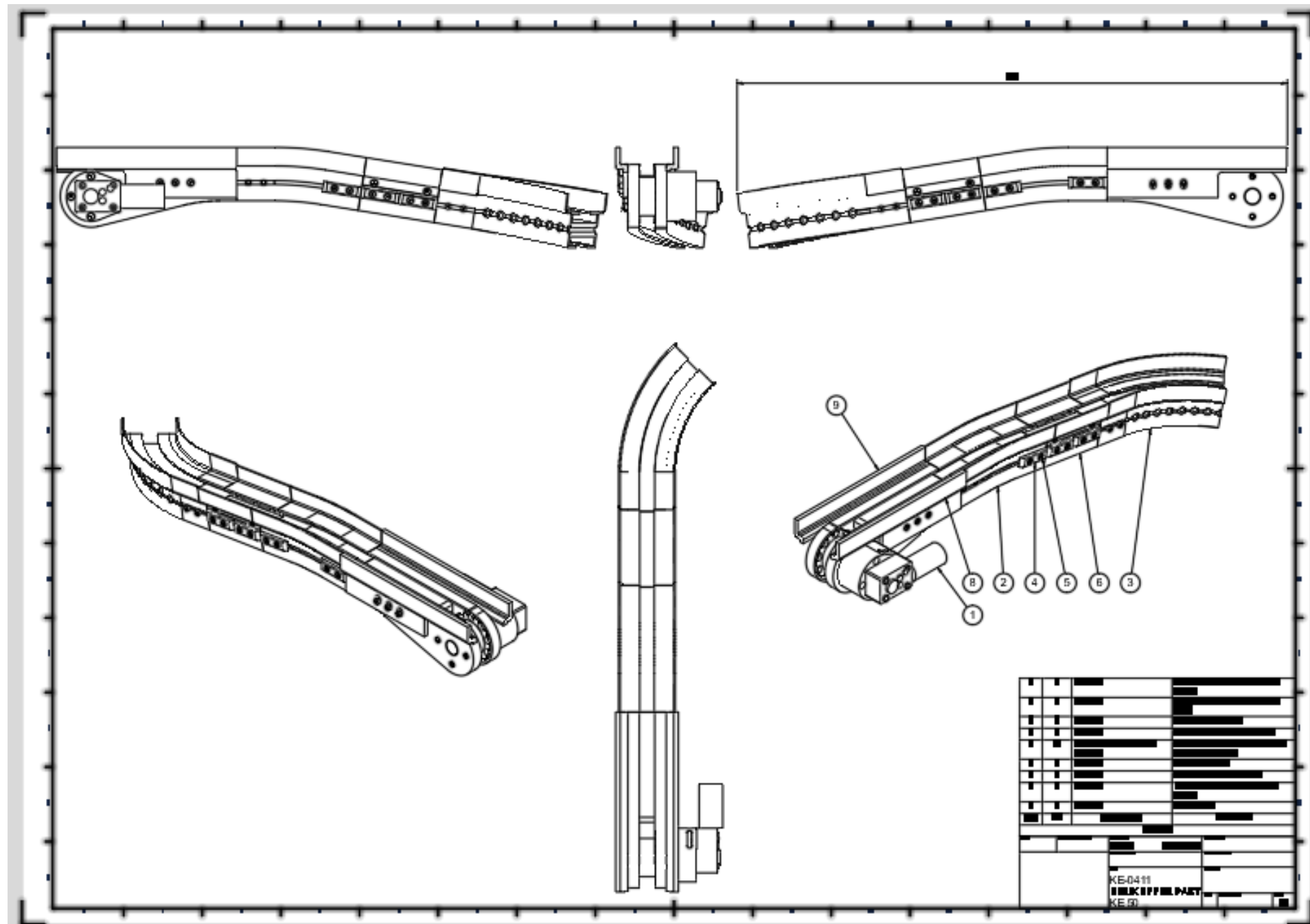
04-Drawings



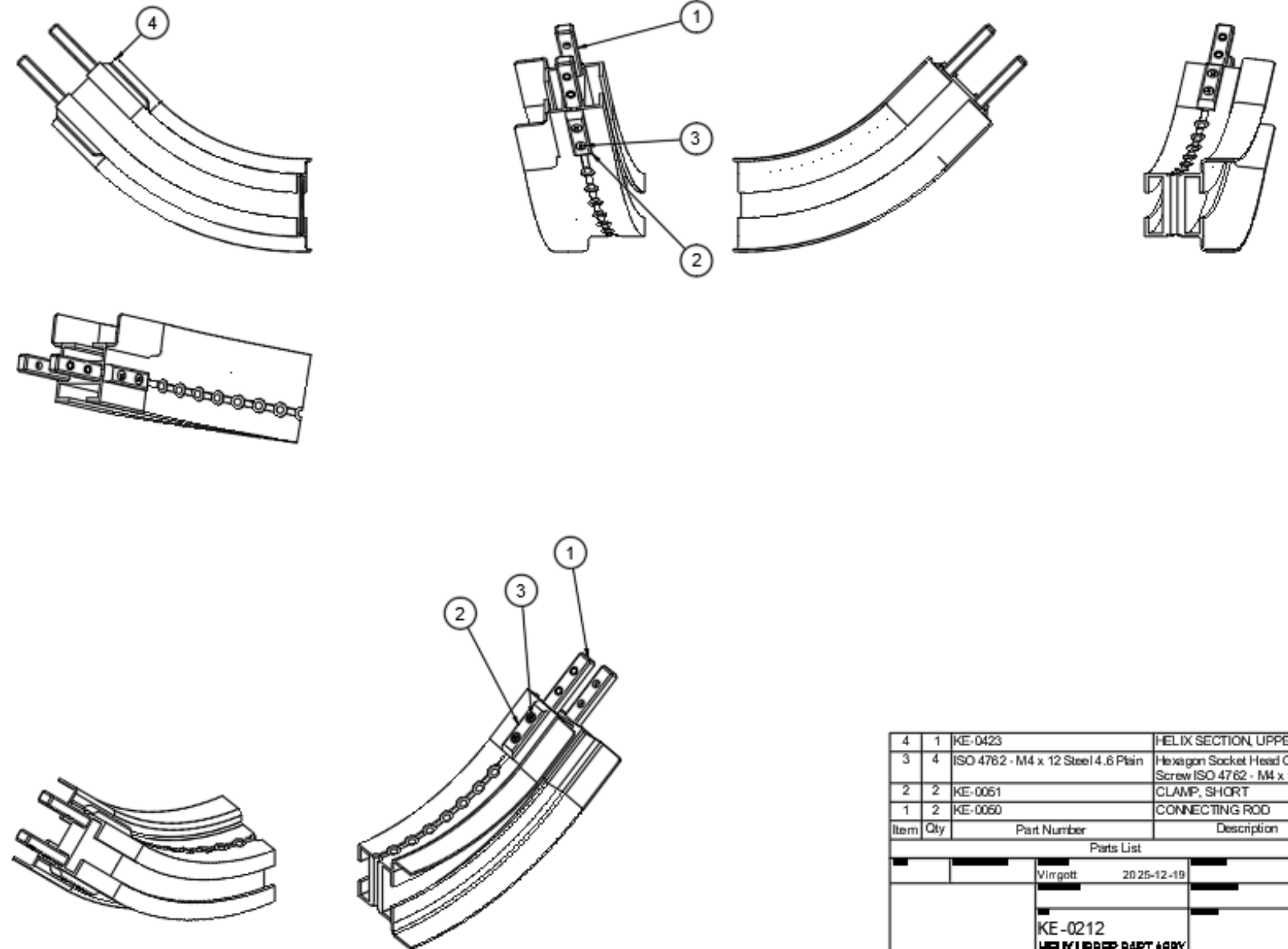
4	1	KE-0058	HINGE
3	1	DA022	3X30mm Stainless Steel Dowel Pin
2	2	KE-0145	FRICTION TOP CLEAT
1	2	KE-0144	CHAIN BODY
Item	Qty	Part Number	Description

Parts List			
		Virgott	2025-11-29
		KE-0149	
		Friction Top Chain	
		KE 50	
			1/1

04-Drawings

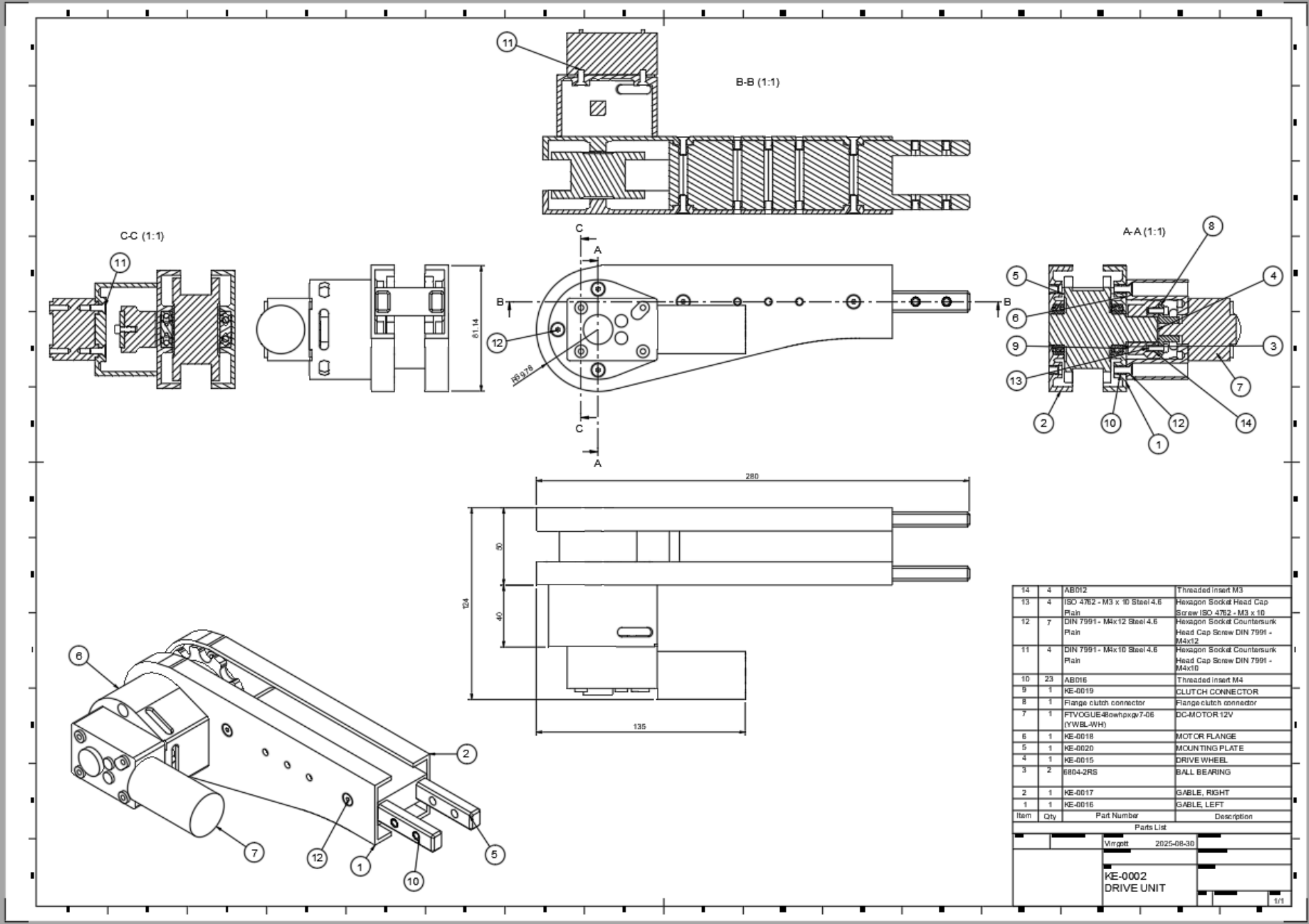


04-Drawings



4	1	KE-0423	HELIX SECTION, UPPER END
3	4	ISO 4762 - M4 x 12 Steel 4.8 Plain	Hexagon Socket Head Cap Screw ISO 4762 - M4 x 12
2	2	KE-0051	CLAMP, SHORT
1	2	KE-0050	CONNECTING ROD
Item	Qty	Part Number	Description
Parts List			
		Virgott	20 25-12-19
		KE-0212	
		HELIX UPPER PART ASBY	
		KE 50	
			1/1

04-Drawings



05-Links to vendors of external parts

It will take a few external. Non-3d-printed parts to build the conveyor.I bought everything on Amazon or Bambulab ´s Makers Supply. Here are the links:



12V DC motor with gearbox, 27rpm:
[Amazon.com: BRINGSMART 12V 12rpm DC Worm Gear Motor 70kg.cm High Torque Self-Locking Reversed Mini Turbine Geared Motor for DIY Robot Rotating Table Door Lock Curtain Machine \(12V 12rpm\): Electronics](#)



Flange clutch connector:
[4Pcs 8mm Flange Coupling Connector, Rigid Guide Model Coupler Accessory, Shaft Axis Fittings for DIY RC Model Motors: Amazon.com: Industrial & Scientific](#)



Filament for conveyor chain:
[PETG HF | Bambu Lab EU Store](#)



Ball Bearings for drive and idler:
[HiPicco 6804-2RS Deep Groove Ball Bearings, ID 20mm x OD 32mm x Width 7mm Double Rubber Sealed Ball Bearing Pre-Lubricated Chrome Steel Z2 Lever 10pcs: Amazon.com: Industrial & Scientific](#)



Ball Bearings for wheel bends:
[Steel Deep Groove Ball Bearings | Bambu Lab EU Store](#)



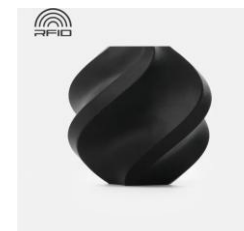
Filament for everything touching the chain:
[PETG - Glitz Grey - add:north](#)



Threaded inserts M3:
[Round Threaded Brass Heat-Insert Nut | Bambu Lab EU Store](#)



Threaded inserts M4:
[Round Threaded Brass Heat-Insert Nut | Bambu Lab EU Store](#)



Filament for everything not touching the chain:
[PETG HF | Bambu Lab EU Store](#)

05-Links to vendors of external parts



Filament for TPU feet:
[TPU 95A HF | Bambu Lab EU Store](#)



Cable connector in battery box:
[Amazon.com: Wago 221-612 Lever-Nuts 10AWG 2 Conductor Compact Wire Connectors 10 PK : Tools & Home Improvement16](#)



M3 cylinder head, insex screw kit:
[M3 Hex Socket Head Cap Screws Bolts and Nuts and Washers Assortment Kit, 801 Pcs Metric Allen Hex Socket Head Cap Screws Bolts, 12.9 Grade Alloy Steel, Fully Threaded, Black Oxide Finish: Amazon.com: Industrial & Scientific](#)



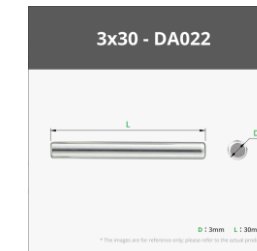
M4 cylinder head, insex screw kit:
[M4 Hex Socket Head Cap Screws Bolts and Nuts and Washers Assortment Kit, 405 Pcs Metric Allen Hex Socket Drive Screws Bolts, 12.9 Grade Alloy Steel, Black Oxide Finish, Fully Threaded: Amazon.com: Industrial & Scientific](#)



M3 countersunk head, insex, screw kit:
[Amazon.com: 260 Pcs M3 Hex Socket Flat Head Countersunk Bolts Screw Nut Assortment Kit- 304 Stainless Steel : Industrial & Scientific](#)



M4 countersunk head, insex, screw kit:
[Hilitchi 180-Piece Metric M4 Hex Socket Flat Head Countersunk Bolts Screw Nut Assortment Kit - 304 Stainless Steel: Amazon.com: Industrial & Scientific](#)



D3 Dowel pin:
[Stainless Steel Dowel Pin | Bambu Lab EU Store](#)